Multimodal Wearable Health Monitoring for Older Adults

Project Progress Report

submitted to

National Institute of Technology Warangal

by

K. Vivek Kumar · 23ECB0A09

Bhanu Pratap · 23ECB0A27

Ujwal Kumar · 23ECB0A16

Under the guidance of

Mohammad Farukh Hashmi



Department of Electronics and Communication Engineering

National Institute of Technology Warangal

November 2026

Approval of the Viva-Voce Board

Project review record

Multimodal Wearable Health Monitoring for Older Adults

Submitted by K. Vivek Kumar · 23ECB0A09, Bhanu Pratap · 23ECB0A27, Ujwal Kumar · 23ECB0A16.

Department of Electronics and Communication Engineering, National Institute of Technology Warangal.

Review date _____

Decision of the board _____

Board member _____ Board member _____

Project guide _____ Chairperson _____

Certificate

For endorsement by the project guide on departmental letterhead.

Project title

Multimodal Wearable Health Monitoring for Older Adults

Team members

K. Vivek Kumar · 23ECB0A09

Bhanu Pratap · 23ECB0A27

Ujwal Kumar · 23ECB0A16

This progress report records the dataset studies, model comparisons and wearable validation work presented for departmental review.

Guide’s signature _____

Mohammad Farukh Hashmi

Department of Electronics and Communication Engineering

National Institute of Technology Warangal

Date _____

Declaration

We submit this interim report on the design and evaluation of a wearable health monitoring system. The four public datasets and the published methods used in the work are identified in the text and references. Experimental results refer to the cohorts and evaluation protocols stated for each module.

We submit the following declaration for signature: the project work described is our own except where another source is acknowledged, and all reported measurements are identified by their source.

K. Vivek Kumar · 23ECB0A09 _____

Bhanu Pratap · 23ECB0A27 _____

Ujwal Kumar · 23ECB0A16 _____

Date _____

Acknowledgements

We thank our project guide, Mohammad Farukh Hashmi, and the Department of Electronics and Communication Engineering at the National Institute of Technology Warangal for supporting this project.

We acknowledge the participants, clinicians and research teams who made PTB-XL, OSASUD, SisFall and WESAD available. Their recordings and annotations made it possible to study four sensing tasks before collecting prospective wearable measurements.

K. Vivek Kumar · 23ECB0A09

Bhanu Pratap · 23ECB0A27

Ujwal Kumar · 23ECB0A16

Abstract

We investigated how far a wearable system for older adults can progress using public physiological recordings before collecting its own hardware data. Four independent pipelines estimate LVH-associated ECG patterns, apnea or hypopnea from optical pulse, simulated falls from inertial motion, and laboratory stress-associated arousal from wrist conductance. Each pipeline uses a separate signal representation, participant-independent evaluation and decision timescale.

For PTB-XL, a single-lead residual CNN achieved patient average precision of 0.437 and detected 105 of 214 positive test ECGs at the wearable threshold. For OSASUD, a multiscale CNN with a Transformer matched 52.2% of respiratory events while generating 68.8 false intervals per hour. For SisFall, a teacher-assisted dual-stream temporal convolutional network averaged 98.61% subject-balanced accuracy over three training seeds. For WESAD, a resting-calibrated, wrist-EDA-only CNN reached mean participant macro F1 of 0.829.

The fall study supplies the clearest evidence for controlled trial recognition. Respiratory alarm burden, single-lead ECG sensitivity, the WESAD laboratory schedule and limited elderly fall observations constrain field use. Paired sensor validation, long-duration daily-activity recordings and timing measurements on the Jetson are the next milestones.

Index terms — wearable sensors, electrocardiography, optical pulse, fall detection, skin conductance, subject-independent evaluation.

Contents

Approval of the Viva-Voce Board	ii
Certificate	iii
Declaration	iv
Acknowledgements	v
Abstract	vi
List of Figures	viii
List of Tables	x
List of Abbreviations	xi
Chapter 1 Introduction	1
Chapter 2 Literature Review	2
Chapter 3 Theoretical Study	3
Chapter 4 Experimental Study	7
Chapter 5 Conclusions and Future Scope of Study	11
References	12
Appendix A Experimental figures	13
Appendix B Reproducibility and validation record	32

List of Figures

A.1 ECG preprocessing — Preprocessing example 13
A.2 ECG preprocessing — Preprocessing fidelity 13
A.3 LVH versus normal — Model comparison 14
A.4 LVH versus normal — Confusion matrices 14
A.5 LVH versus normal — ROC, precision–recall and calibration 14
A.6 LVH versus normal — Learning curves 14
A.7 LVH versus normal — Loss curves 15
A.8 LVH versus normal — Runtime comparison 15
A.9 LVH versus normal — Age subgroups 15
A.10 LVH versus normal — Coverage and subgroup checks 15
A.11 LVH versus rest — Model comparison 16
A.12 LVH versus rest — Confusion matrices 16
A.13 LVH versus rest — ROC, precision–recall and calibration 16
A.14 LVH versus rest — Learning curves 16
A.15 LVH versus rest — Loss curves 17
A.16 LVH versus rest — Runtime comparison 17
A.17 LVH versus rest — Age subgroups 17
A.18 LVH versus rest — Coverage and subgroup checks 17
A.19 ECG follow-up — Ablation 18
A.20 ECG follow-up — Heldout curves 18
A.21 Optical pulse — Models 19
A.22 Optical pulse — Patient performance 19
A.23 Optical pulse — Training 19
A.24 Optical pulse — Ablations 20
A.25 Optical pulse — Precision–recall and calibration 20
A.26 Optical pulse — Respiratory subtype confusion 20
A.27 PPG follow-up — Models 21
A.28 PPG follow-up — Alert budgets 21

List of Figures (continued)

A.29 IMU — Dataset profile 22
A.30 IMU — Native signals 22
A.31 IMU — Gravity components 22
A.32 IMU — Filter peak retention 23
A.33 IMU — Ablations 23
A.34 IMU — Model comparison 24
A.35 IMU — Three primary model comparisons 24
A.36 IMU — Confusion 25
A.37 IMU — Discrimination calibration 25
A.38 IMU — Neural training 25
A.39 IMU — Training-seed sensitivity 26
A.40 IMU — Transfer 26
A.41 IMU — Secondary transfer controls 26
A.42 IMU — Transfer of distilled DM-TCN 26
A.43 EDA — Cohort conditions 27
A.44 EDA — Protocol examples 27
A.45 EDA — Decomposition comparison 27
A.46 EDA — Eda only architectures 28
A.47 EDA — Architectures 28
A.48 EDA — Component ablations 29
A.49 EDA — Protocol challenge 29
A.50 EDA — Export candidate curves 29
A.51 EDA — Zero-calibration participant decisions 30
A.52 EDA — Zero-calibration discrimination curves 30
A.53 EDA — Rest-calibrated participant decisions 30
A.54 EDA — Rest-calibrated discrimination curves 30
A.55 EDA — Learned decomposition 31
A.56 EDA — Learning curves 31

List of Tables

1.1 Wearable inputs and study endpoints 1
4.1 Lead II models across two negative-class definitions 7
4.2 Respiratory models at their event thresholds 8
4.3 Selected primary comparisons at 200 Hz 9
4.4 Participant means on matched EDA windows 10
B.1 Evidence needed to verify the selected pipelines 32

List of Abbreviations

ADC Analog-to-digital converter

ADL Activity of daily living

AP Average precision

AUROC Area under the receiver operating characteristic curve

BA Balanced accuracy

CNN Convolutional neural network

ECG Electrocardiogram

EDA / GSR Electrodermal activity / galvanic skin response

GRU Gated recurrent unit

IMU Inertial measurement unit

IQR / MAD Interquartile range / median absolute deviation

LOSO Leave-one-subject-out evaluation

LVH Left ventricular hypertrophy

MIL Multiple instance learning

ONNX Open Neural Network Exchange

PPG Photoplethysmography

RMS Root mean square

TCN Temporal convolutional network

Chapter 1

Introduction

1.1 Measurement objectives

A waist-mounted sensor can register a sudden rotation within milliseconds, while a useful change in skin conductance may unfold over a minute. Our system keeps a separate clock, quality decision and model for each measurement. A reviewer can trace every prediction back to its observation interval and compare it against a suitable reference.

Table 1.1 Wearable inputs and study endpoints

Module	Sensor	Research output
ECG	BioAmp EXG Pill	LVH-associated Lead II pattern
PPG	MAX30102	Apnea or hypopnea interval
IMU	Waist MPU-9250	Fall versus daily activity
EDA	Wrist GSR sensor	Baseline versus stress condition

1.2 Scope of the progress review

The offline work covers source audits, participant splits, signal preparation, training, calibration and held-out evaluation. Live processing uses acquisition buffers, module-specific inference and timestamped outputs. Model deployment and clinical usefulness require additional measurements from the intended sensors, placement and wearer population.

The local records preserve the public source data. Fitted parameters, predictions, exports and performance graphs are stored separately so that a subsequent run can be compared with the original experiment.

Chapter 2

Literature Review

2.1 The reference defines the learning task

Wagner et al. [1] released ten-second, twelve-lead ECGs with diagnostic statements. PTB-XL benchmarks [2] established patient-independent evaluation for ECG interpretation. A wearable Lead II approximation removes the spatial information provided by the other leads. The project therefore asks how well an ECG statement can be recovered with that reduced input.

OSASUD [3] associates bedside monitoring with respiratory reference signals in a stroke-unit cohort. Pulse amplitude and heart-beat timing provide indirect evidence about breathing. Removing oximetry and respiratory reference channels makes a one-channel optical experiment relevant to the wearable, but also changes the information available for distinguishing apnea subtypes.

Sucerquia et al. [4] collected labeled fall and daily-activity trials using waist sensors. Trial labels support whole-record evaluation. Localizing an exact impact requires additional annotation. We treat each trial as a collection of candidate windows and measure repeated alarms during daily activities separately.

Schmidt et al. [5] introduced WESAD with synchronized laboratory conditions and wrist and chest signals. Skin conductance decomposition separates slow and rapid components, as illustrated by cvxEDA [6]. Our wrist-only design tests a causal decomposition and a personal resting reference under leave-one-person-out evaluation.

2.2 Gap addressed

The common research question is how signal processing and validation change when a deployed device has fewer channels, a continuous clock and a limited opportunity to calibrate. The experiments measure the effects of those constraints separately for each modality.

Chapter 3

Theoretical Study

3.1 ECG voltage and morphology

The BioAmp measures a voltage difference. The model learns which voltage shapes and amplitudes across ten seconds of one lead predict the LVH diagnostic statement used in PTB-XL. The reference is a clinician-interpreted ECG statement. Anatomical confirmation by echocardiography falls outside this experiment.

Construct the learning target

Read the calibrated 500 Hz PTB-XL waveform and select Lead II from the twelve recorded leads. An ECG with the LVH statement is positive. Other eligible ECGs, including those with other abnormalities, are negative. Excluding a contradictory LVH/NORM record leaves 21,798 ECGs. Keep patient identity for grouping and official folds for splitting.

Preserve measured voltage and timing

Express the lead in millivolts. Resample each complete 10 s recording from 500 to 360 Hz with a polyphase anti-aliasing FIR filter (Kaiser window, $\beta = 5$). A calibrated native 360 Hz input already has the required 3,600 samples. The selected minimal pipeline retains absolute amplitude. Per-record standardization would erase voltage differences the detector can use.

Recognize waveform shapes

A 1D convolution groups 18 samples (50 ms) into each of 200 time positions with 32 learned features. Two residual blocks apply parallel kernels of 3 and 7 positions with dilations 1 and 2 respectively. Short kernels describe local shapes while longer, spaced kernels combine them over more of a heartbeat. Each block concatenates its two paths, mixes them and adds its input back. Temporal mean and maximum pooling retain sustained patterns and strong local responses. The resulting 64 values feed a 48-unit layer and one raw score, called a logit.

Absolute voltage reaches the classifier through a single channel. Calibration and thresholding operate after the learned waveform score.

Optical pulse and breathing

3.2 Respiratory modulation

A pulse waveform contains both individual pulses and slower changes in their level, size and spacing. The model combines these to estimate whether a clinician marked apnea or hypopnea at a given second. A pulse sensor measures blood-volume changes, so the evidence for an airway event is indirect.

Retain the pulse shape

For each adjacent pair of samples, keep their mean and shrink the finest Haar detail using a robust within-second noise estimate. Reconstruct the pair. This reduces rapid variation while preserving the local mean. Band-pass at 0.45–5.5 Hz and take 16 evenly spaced pulse-shape values per second, scaled by a slowly tracked pulse range.

Retain slower changes and describe quality

From a separate 6 Hz low-pass trace compute median level and robust pulse range (90th minus 10th percentile). Use trailing 45 s exponential references for baseline, amplitude and beat-interval changes. Estimate the beat interval from accepted upward crossings. Filter the three changes at 0.08–0.38 Hz and track their energy over an 8 s exponential timescale. Pulse RMS and beat interval complete 11 modulation values. Four quality values describe high-frequency residual, repeated samples, time since a reliable beat and dwell at an observed extreme. These help distinguish an optical plateau from a physiological change.

Build and encode the context

Each second now has 31 values (16 shape + 11 modulation + 4 quality). Apply training-patient medians and interquartile ranges, then inverse-hyperbolic-sine compression to limit the influence of extreme values. Group 30 consecutive seconds and advance by 5 s. Project each view to width 32, join the views and apply a residual temporal convolution. Parallel 3 s and 7 s convolutions capture local changes. Positional encoding and one four-head Transformer layer relate those changes across the completed 30 s context. A shared head reads the last five positions separately. This preserves event boundaries that a single label for the whole window would blur.

Each five-second update assigns a separate score to each of the final five seconds. A transition within that block therefore survives training.

Impact and body motion

3.3 Weak supervision

Acceleration describes impact and changes in body motion. Angular velocity describes rotation. A short burst in either can also occur during ordinary activity. The network estimates which combinations of both signals make a recording resemble a simulated fall.

Preserve impacts while estimating body motion

IPMV (Impact-Preserving Multi-View preprocessing) keeps native-rate magnitudes on a direct path. For each 50 ms token, preserve short extrema and the largest absolute one-sample magnitude change. A parallel causal second-order, 30 Hz adaptive filter passes more raw residual during large changes. A 0.75 Hz filter estimates the gravity vector. Subtract it to estimate dynamic acceleration and project motion and rotation parallel and perpendicular to gravity. These measures survive a common fixed rotation of the axes. Rapid body rotation can still make the gravity estimate lag.

Represent several timescales

Apply a three-level causal Daubechies-2 wavelet decomposition to acceleration and gyro magnitudes. Summarize each detail band and the remaining coarse band by RMS per token. Combine these with impact and gravity-relative measures to form 24 values every 50 ms. Standardize by training-subject means and standard deviations and apply inverse-hyperbolic-sine compression. Feed 40 tokens (2 s) at a time with a 1 s hop. Windows have equal length and no padding. A duration-only control classified every trial correctly, showing why full file length would be a misleading input.

Learn from trial labels without inventing an impact time

Treat each recording as a bag of windows. During training, sample eight positions and apply the trial loss to the maximum window logit. A training-only XGBoost teacher starts with high-acceleration candidate windows, then refines its candidates using trial labels. Its softened window scores (temperature 2) guide the neural network alongside the trial loss. At evaluation, scan all complete windows and use the strongest evidence for the trial. Attention weights and high-scoring windows identify model evidence, with no reference annotation for the exact impact instant.

Feature-wise gating combines the acceleration and rotation encoders. Attention pools the completed window before a scalar fall-evidence score is produced.

Wrist conductance and personal calibration

3.4 Tonic and phasic responses

Skin conductance has a slowly varying level (tonic) and shorter responses (phasic). This model compares both with the wearer's own resting conductance, then estimates whether a minute of wrist EDA resembles WESAD's baseline or stress condition. It requires one wrist EDA sensor and a designated 120 s resting capture. The waist IMU serves the fall model independently.

Assess contact and clean continuously

Flag nonfinite or out-of-range values (research plausibility range 0–100 µS). Use the last valid value for short invalid stretches. Compare changes with a trailing 30 s median and median absolute deviation (MAD). Step reversals, sustained flatness and near-zero conductance indicate possible artifacts. Attenuate suspect changes and apply a causal two-pole 0.65 Hz low-pass. Preserve raw conductance as a separate input. A gap of 5 s resets filter state and excludes intersecting windows. Exclude windows with over 20% missing observations. All quality estimates for this configuration come from EDA.

Separate slow level from rapid responses

In the project's MC-SAED decomposition, a quality-weighted tonic tracker follows downward relaxation faster than upward excursions. This leaves room for positive pulses above baseline. Smooth the positive remainder to form the phasic estimate. During the designated rest, require at least 384 usable quarter-second samples. For each component, subtract its resting median and divide by 1.4826 × resting MAD, with scale floors of 0.05 µS for tonic and 0.02 µS for phasic. Keep both absolute and rest-relative channels. The resting reference is the only personalized measurement required.

Pass 12 signals to a compact CNN

Each window has 240 samples of raw, cleaned, tonic, phasic, relative tonic, relative phasic, tonic slope, phasic slope, phasic energy, quality, artifact and missingness. Fit numeric medians and interquartile ranges on training subjects. Keep quality indicators on their original scales. A convolution with width and stride 4 embeds each second into 16 features. Two temporal convolutions with kernels of 5 and 3 seconds and GELU activations learn patterns across adjacent seconds. Average the 60 output positions, then apply layer normalization, dropout during training and a two-class linear head. The model has 2,930 trainable parameters.

The waist IMU supplies the fall detector independently. All cleaning, reference estimates and decisions in the selected arousal model use wrist EDA alone.

Chapter 4

Experimental Study

4.1 ECG model comparison

The first comparison trained five seeds per architecture on official patient folds. LVH versus normal restricts negatives to normal ECGs. LVH versus rest retains other diagnostic abnormalities and better reflects the intended screening task. Mean test record-level results are reported below [7].

Table 4.1 Lead II models across two negative-class definitions

Negatives	Architecture	AUROC	AP	Sens.	Spec.
normal	CNN	0.933	0.884	0.822	0.908
normal	CNN + SSM	0.900	0.837	0.763	0.905
normal	CNN + attention	0.910	0.851	0.771	0.912
rest	CNN	0.811	0.402	0.520	0.892
rest	CNN + SSM	0.792	0.378	0.475	0.895
rest	CNN + attention	0.787	0.378	0.477	0.895

Single-channel follow-up

The later study trained challengers on folds 1–6, ranked them on folds 7–8, refit on folds 1–8 and calibrated on fold 9. On the previously studied fold 10, the selected minimal residual CNN had patient AP 0.437, AUROC 0.804 and F1 0.429. It detected 104 of 202 positive patients at 89.5% specificity.

At the separate record threshold of 0.18572, it detected 105 of 214 positive ECGs and produced 207 false positives among 1,984 negatives. Twelve-lead teaching did not establish an improvement in a matched student comparison. This result favors preserving physical voltage and measuring single-lead limitations directly.

4.2 Patient-held-out comparison

Each fold used 18 training, six validation and six test patients. Of 961,357 labeled seconds, 821,545 were scored after unusable optical packets and context warm-up. The table compares 30-patient mean AP and pooled interval performance at each model's validation-selected operating threshold [8].

Table 4.2 Respiratory models at their event thresholds

Model	AP	Recall	Event F1	False/h
CNN + Transformer	0.238	0.522	0.252	68.8
TCN + GRU	0.234	0.624	0.281	73.8
Wavelet + state space	0.228	0.603	0.266	76.9
Dual TCN + attention	0.229	0.601	0.260	79.1
Binary TCN + GRU	0.242	0.576	0.276	68.2
Two scales, hard	0.247	0.463	0.257	56.2
Two scales, fraction	0.239	0.476	0.272	53.2
Two scales, auxiliary	0.248	0.451	0.264	51.6

The CNN + Transformer remained the overall validation choice under the AP tolerance and event-F1 tie-break. It matched 3,125 of 5,985 reference intervals. The binary two-scale hard-label model reduced the burden to 56.2 false intervals per hour at 46.3% recall. At a low-burden operating point the original model reached only 0.40% recall. These trade-offs rule out caregiver alerts at the measured operating points.

Restoring slow modulation and quality to morphology improved AP by 0.041 in a matched control. The interval-matching criterion required intersection over union of at least 0.30 and five consecutive positive seconds.

Fall trial classification

4.3 Subject independence and repeated training

Five conflicting identities were excluded. The 4,500 retained trials comprised 1,798 simulated falls and 2,702 daily activities from 38 people. Five outer participant groups assessed fresh refits. The primary screen and repeated-seed follow-up answer different questions [9].

Table 4.3 Selected primary comparisons at 200 Hz

Model	Subject BA	Recall	F1	False/h
InceptionTime	0.985	0.982	0.981	2.41
XGBoost	0.985	0.987	0.980	3.79
Random forest	0.984	0.985	0.980	3.52
SVM	0.983	0.983	0.979	3.93
Dual CNN–TCN	0.927	0.862	0.919	1.93
1D CNN	0.780	0.620	0.725	14.28

The teacher-assisted dual-stream TCN achieved 98.20%, 98.58% and 99.04% subject-balanced accuracy on the same splits with three seeds. Its false alert rates were 2.21, 2.34 and 3.10 per scripted daily-activity hour. Training the same backbone using only trial loss produced 99.04%, 82.62% and 99.04%. The teacher mainly stabilized one poor fit.

A duration-only tree classified every trial correctly. Fixed unpadded windows reduce access to this shortcut. The replay counts threshold excursions with a five-second refractory period and two below-threshold windows to re-arm. Only one older participant contributed simulated falls, limiting conclusions about elderly fall sensitivity.

4.4 Wrist sensor constraint and calibration

The 15 released participants supplied 1,008 baseline and 628 stress windows. Within each leave-one-subject-out fold, 12 participants trained the development model and two selected the epoch and temperature. A refit used all 14 development participants. Every model below uses wrist EDA alone [10].

Table 4.4 Participant means on matched EDA windows

Model	Rest	Macro F1	Sens.	Spec.	AUROC
CNN	Yes	0.829	0.823	0.882	0.962
CNN	No	0.735	0.748	0.791	0.932
LSTM	Yes	0.806	0.818	0.852	0.955
CNN–BiLSTM	Yes	0.821	0.791	0.900	0.952
TCN attention	Yes	0.832	0.827	0.880	0.966
Feature hybrid	No	0.777	0.805	0.805	0.944

The calibrated CNN ranked first by inner-validation macro F1 (0.869) among the eligible EDA-only neural candidates. Its held-out macro F1 was 0.829 (participant interval 0.734–0.918). The same CNN trained without a personal reference reached 0.735. The added reference requires 120 seconds of designated rest and at least 384 usable quarter-second ticks.

An elapsed-time-only control reached 0.999 macro F1, showing how strongly the laboratory schedule predicts the condition. The selected CNN receives no session clock. Its score describes laboratory arousal and still needs external validation against exertion, movement and electrode changes.

Chapter 5

Conclusions and Future Scope of Study

5.1 Progress at the review date

All four modules have undergone participant-independent evaluation. ECG and PPG have single-channel research exports. The six-axis fall network has a reference FP32 export. Subject-fold checkpoints support the selected EDA-only CNN, while existing full-cohort EDA exports expect additional sensors. A matching EDA-only refit and export is an outstanding integration task.

Offline-to-runtime checks reproduced ECG logits within 0.000001 on 59 source records. PPG streaming replay reproduced reference probabilities on three real traces. The IMU FP32 graph reproduced reference logits within 0.000002 on 256 windows. These checks establish numerical agreement under replay. Physical sensor transfer and long-duration field performance remain unmeasured.

5.2 Priorities for the next phase

First, measure sensor bias, gain, orientation, contact and timing on the actual boards. Confirm Lead II polarity and input-referred voltage, select the optical channel with a paired reference, bench-calibrate the waist IMU and characterize wrist GSR in microsiemens.

Second, collect protocol-approved paired recordings with person-independent partitions. Long-duration ADL exposure is especially important for fall alarms. Respiratory evaluation requires independently timed reference events and coverage accounting. Wrist arousal needs activity and context controls outside WESAD.

Third, measure end-to-end latency, memory, power, dropped packets and recovery on Jetson Nano. Acquisition delay and window accumulation must be included in addition to model execution time. Each signal should retain its own availability flag and observation interval.

References

[1] Wagner, P. et al. PTB-XL, a large publicly available electrocardiography dataset. Scientific Data 7, 154 (2020). doi:10.1038/s41597-020-0495-6.

[2] Strodthoff, N. et al. Deep Learning for ECG Analysis: Benchmarks and Insights from PTB-XL. IEEE Journal of Biomedical and Health Informatics 25, 1519–1528 (2021). doi:10.1109/JBHI.2020.3022989.

[3] OSASUD data descriptor. A dataset of stroke unit recordings for the detection of obstructive sleep apnea syndrome. Scientific Data (2022). doi:10.1038/s41597-022-01272-y.

[4] Sucerquia, A., López, J. D. and Vargas-Bonilla, J. F. SisFall: A Fall and Movement Dataset. Sensors 17, 198 (2017). doi:10.3390/s17010198.

[5] Schmidt, P. et al. Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection. ICMI (2018). doi:10.1145/3242969.3242985.

[6] Greco, A. et al. cvxEDA: A Convex Optimization Approach to Electrodermal Activity Processing. IEEE Transactions on Biomedical Engineering 63, 797–804 (2016). doi:10.1109/TBME.2015.2474131.

[7] Project ECG experiments. PTB-XL Lead II comparison and LVH distillation reports. Research results directories 20261105_lvh_comparison and lvh_distill_v1.

[8] Project optical experiments. OSASUD respiratory-event study and binary two-time-scale follow-up. Research results directories osasud_v1 and osasud_binary_dual_rate_v1.

[9] Project inertial experiments. SisFall subject-held-out study, repeated seeds and transfer reports. Research results directory sisfall_v1.

[10] Project wrist EDA experiments. WESAD subject-independent comparison, matched-window controls and verification. Research results directory wesad_loso_v1.

Waveform preservation and distortion

The example and fidelity audit belong to the initial five-seed comparison. The later minimal pipeline retains calibrated voltage and uses a different selection experiment.

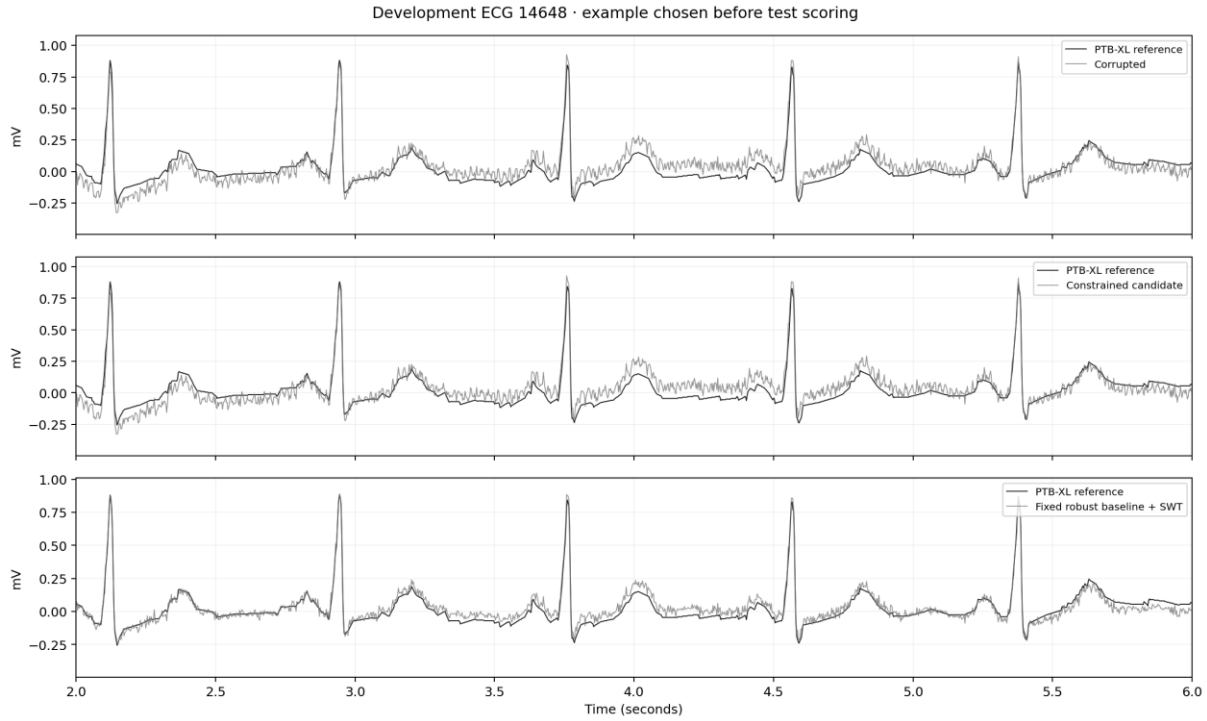


Figure A.1 Preprocessing example

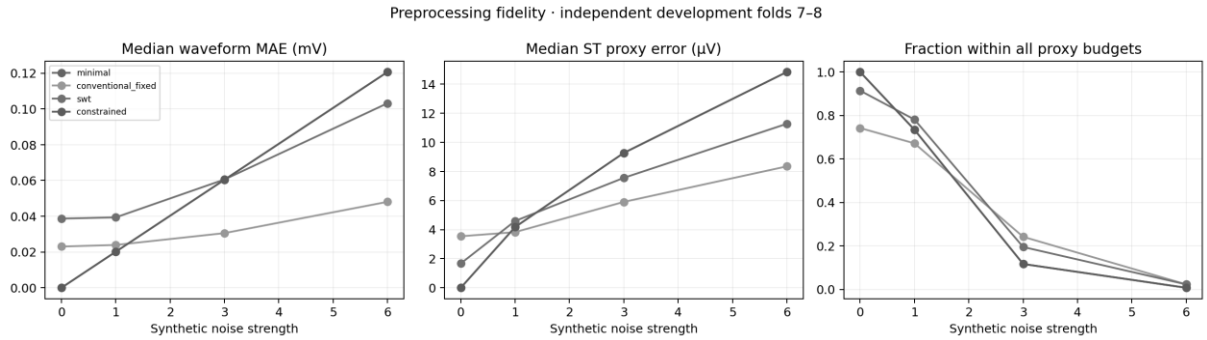


Figure A.2 Preprocessing fidelity

Appendix A · LVH versus normal

Model performance and learning

The comparison uses five independent training seeds. Read discrimination curves alongside error counts at the fold-9 operating threshold.

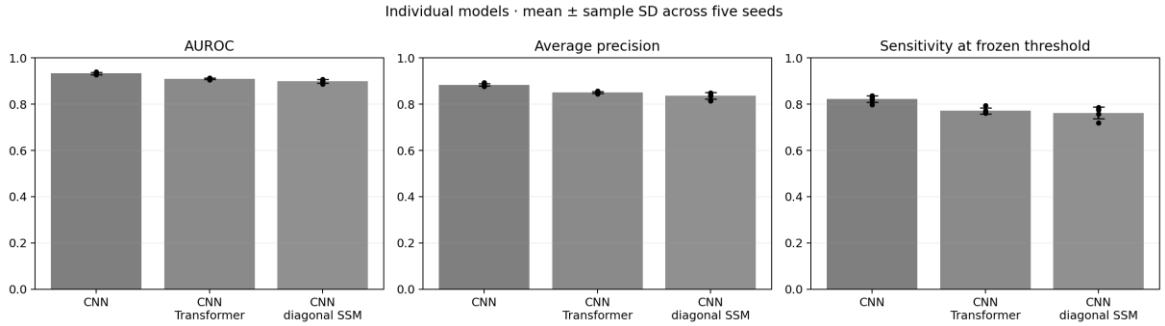


Figure A.3 Model comparison

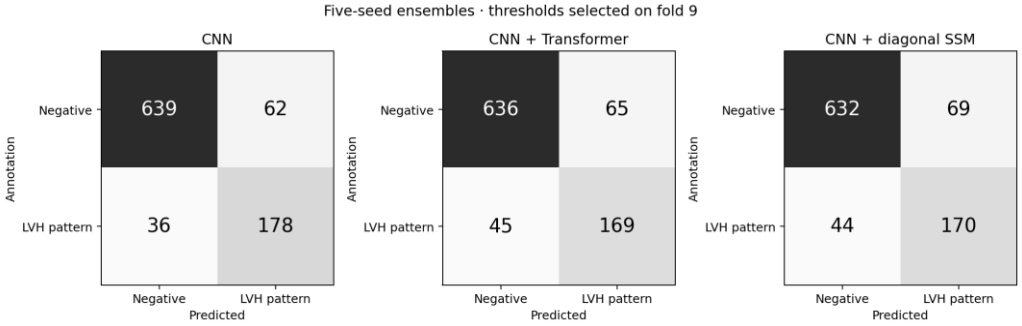


Figure A.4 Confusion matrices

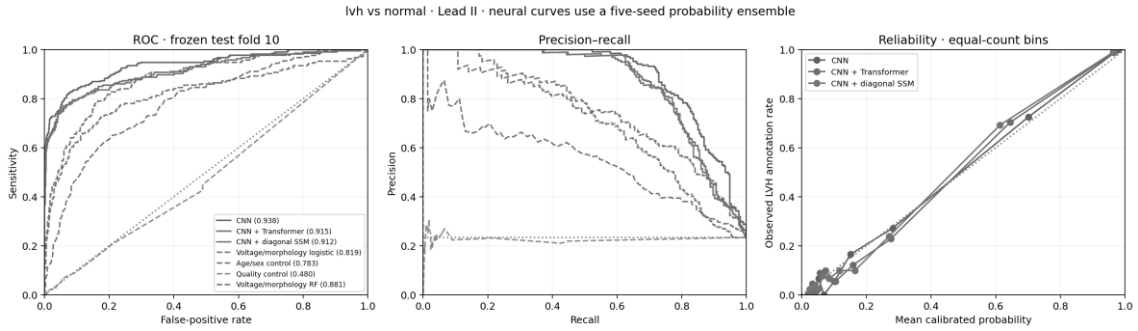


Figure A.5 ROC, precision–recall and calibration

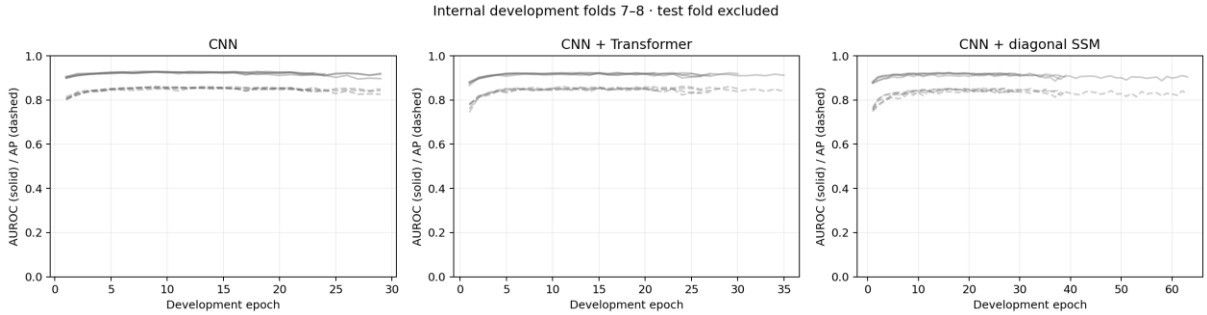


Figure A.6 Learning curves

Training loss, runtime and subgroups

Runtime describes the benchmark host. Subgroup estimates depend on coverage, positive case counts and the negative-class definition.

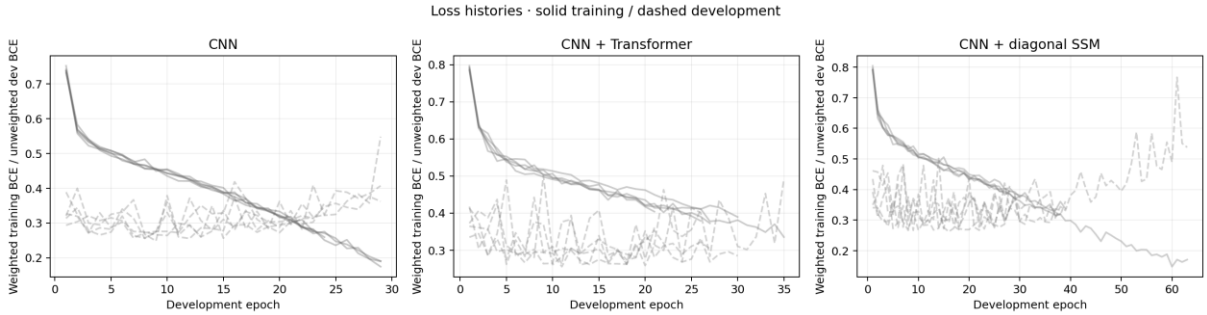


Figure A.7 Loss curves

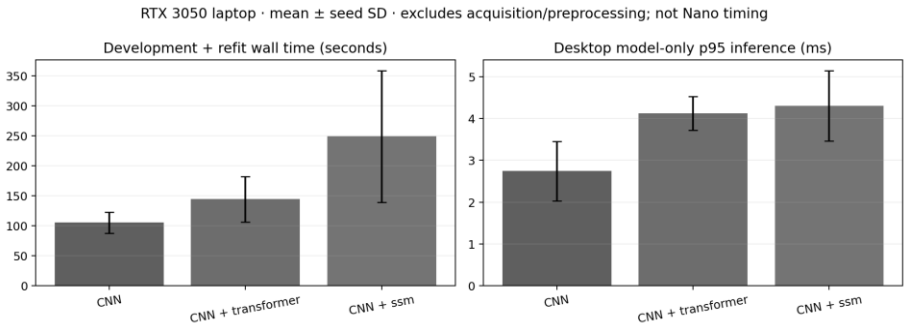


Figure A.8 Runtime comparison

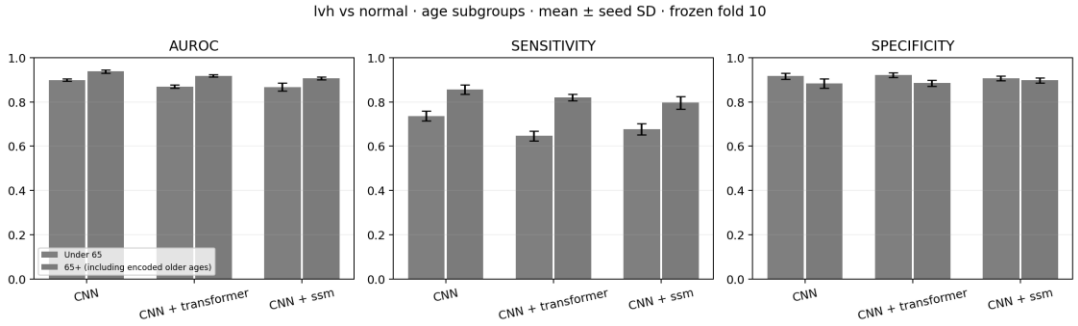


Figure A.9 Age subgroups

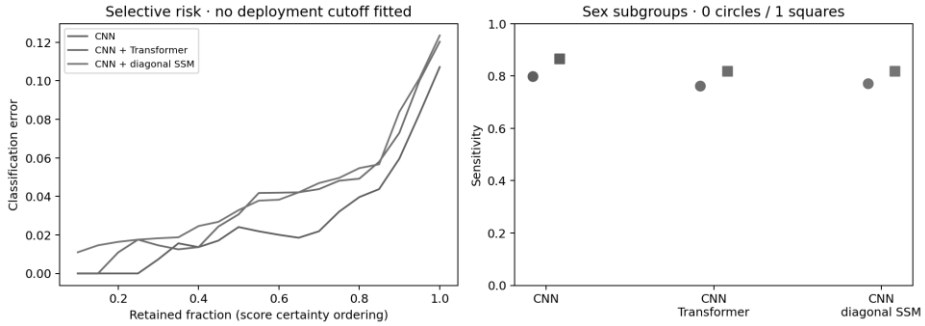


Figure A.10 Coverage and subgroup checks

Model performance and learning

The comparison uses five independent training seeds. Read discrimination curves alongside error counts at the fold-9 operating threshold.

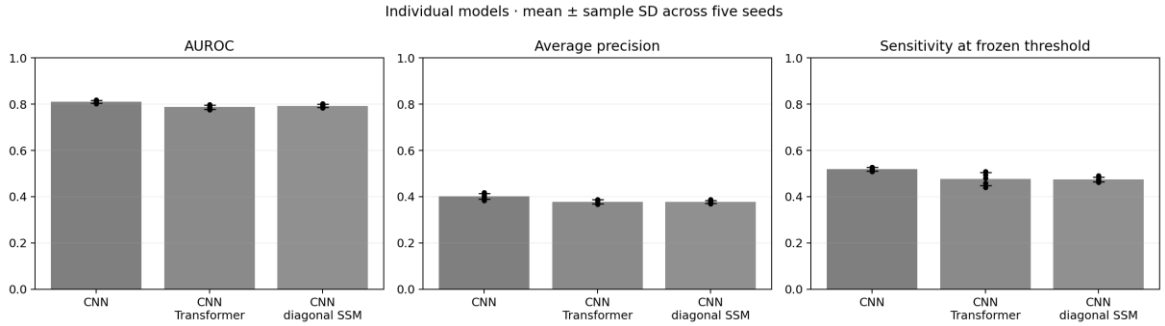


Figure A.11 Model comparison

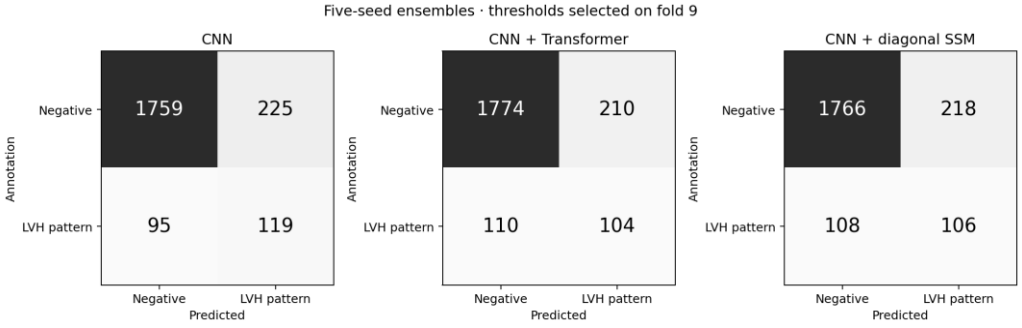


Figure A.12 Confusion matrices

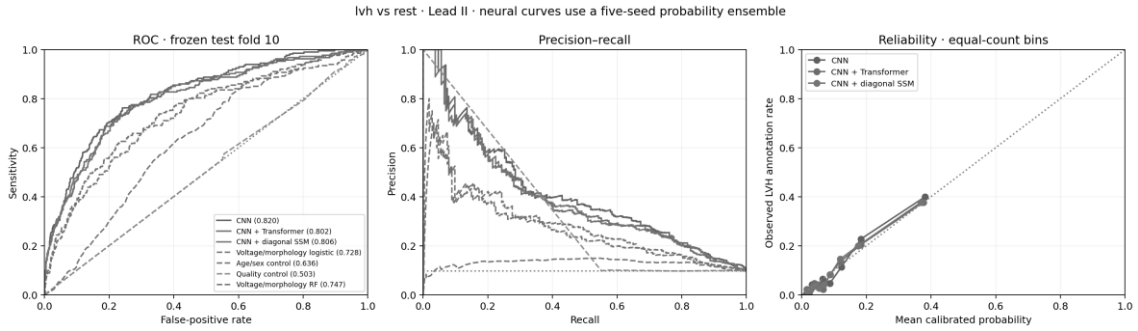


Figure A.13 ROC, precision–recall and calibration

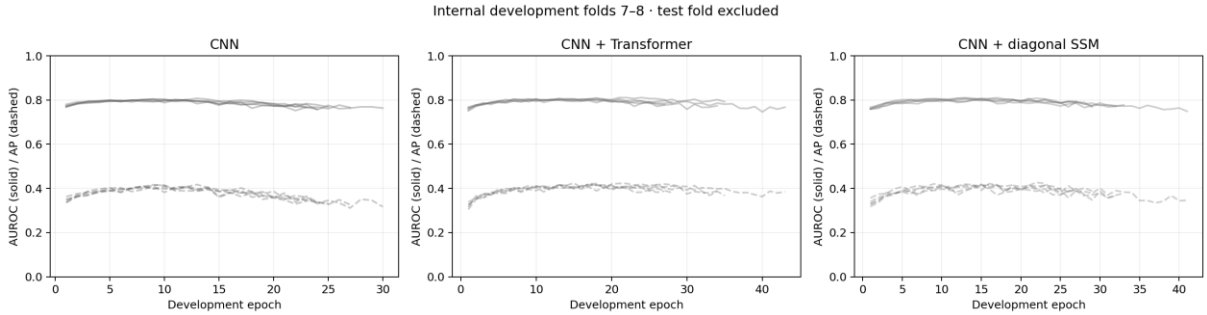


Figure A.14 Learning curves

Training loss, runtime and subgroups

Runtime describes the benchmark host. Subgroup estimates depend on coverage, positive case counts and the negative-class definition.



Figure A.15 Loss curves

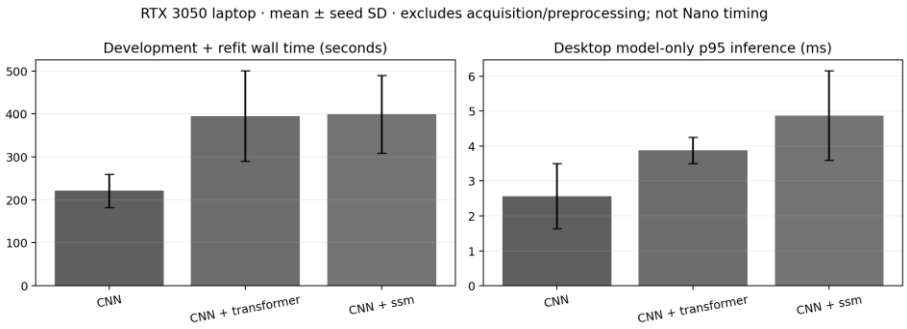


Figure A.16 Runtime comparison

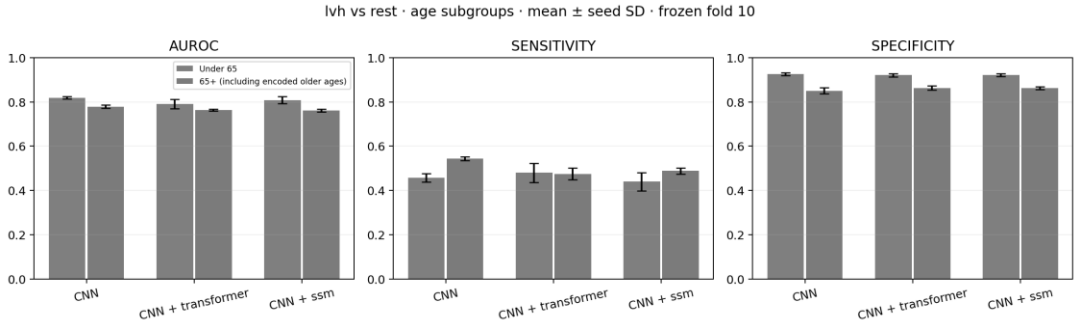


Figure A.17 Age subgroups

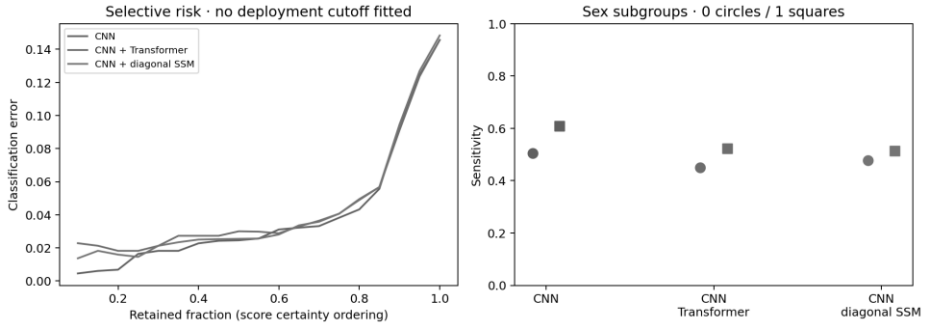


Figure A.18 Coverage and subgroup checks

Ablation and held-out curves

Read the twelve-lead teacher separately from the single-channel configurations. The wearable result comes from the minimal preprocessing model.

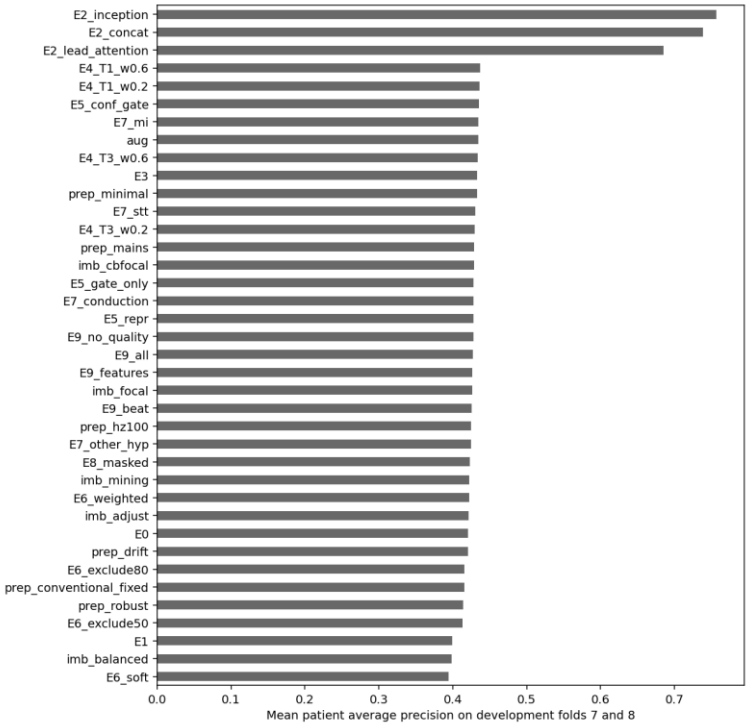


Figure A.19 Ablation

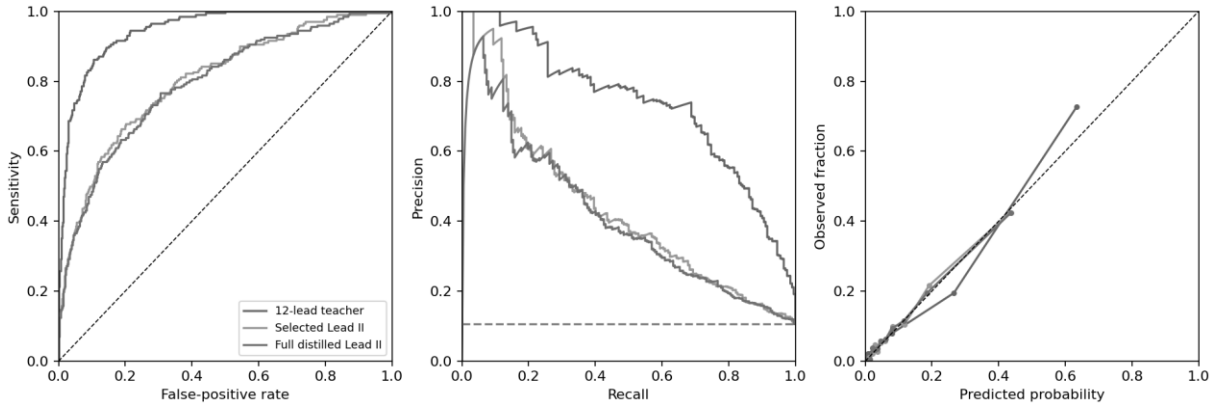


Figure A.20 Heldout curves

Architectures and patient variation

Patients contribute equally to the primary AP statistic. The distribution shows how strongly an average can conceal performance differences across individuals.

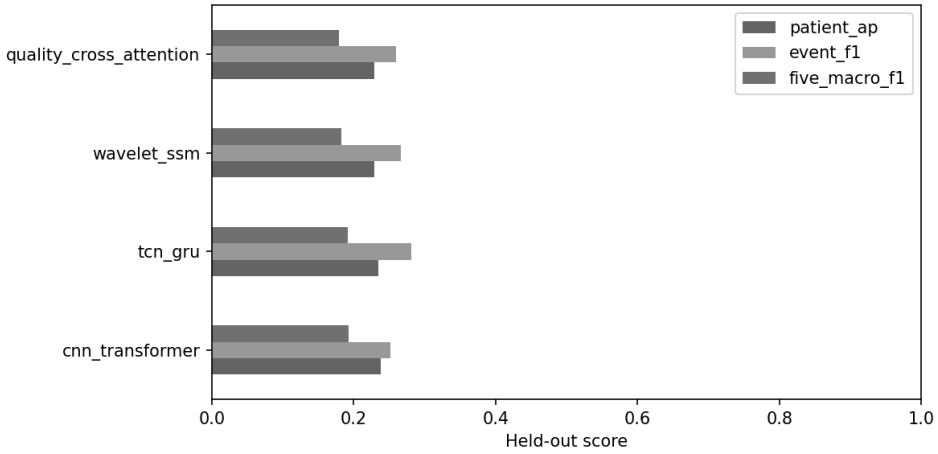


Figure A.21 Models

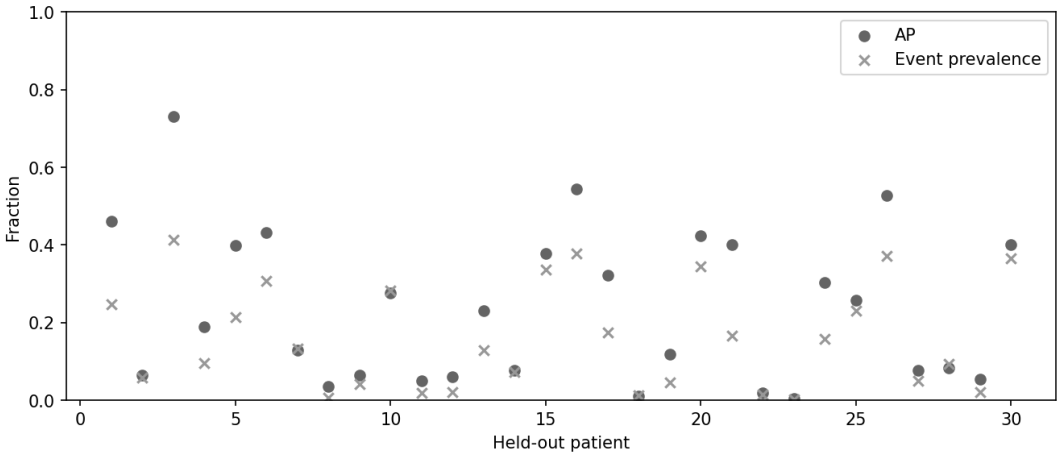


Figure A.22 Patient performance

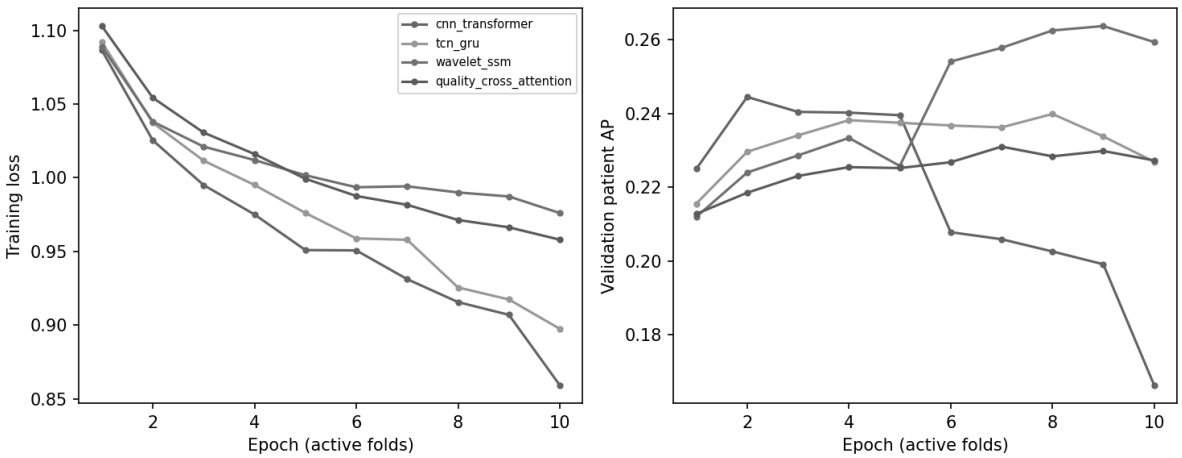


Figure A.23 Training

Feature controls and event subtypes

The primary decision joins apnea and hypopnea into one positive class. The subtype matrix shows why reliable subtype identification remains an open problem.

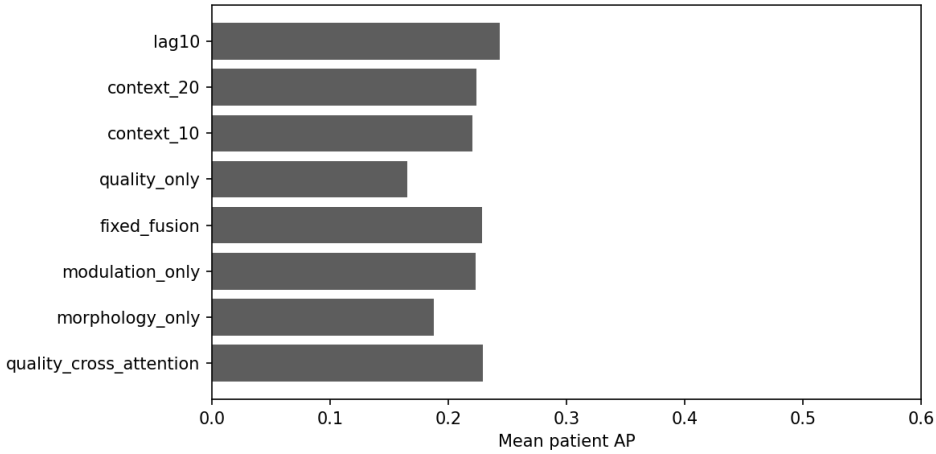


Figure A.24 Ablations

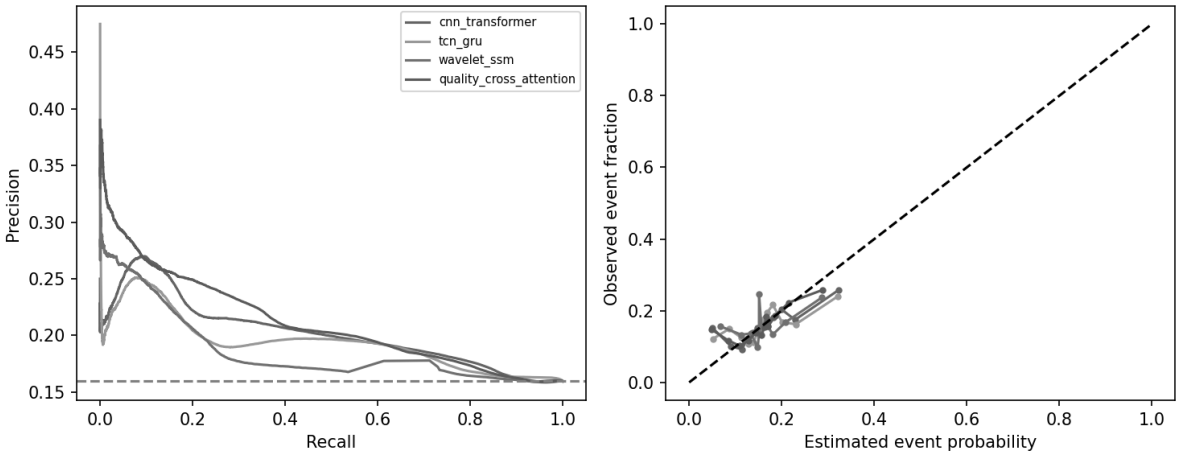


Figure A.25 Precision–recall and calibration

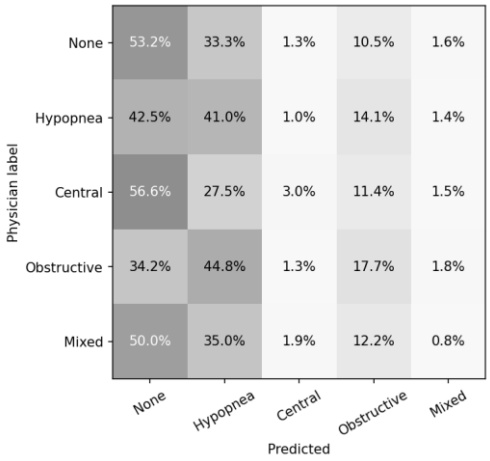


Figure A.26 Respiratory subtype confusion

Two-time-scale models and alarm budgets

The follow-up used the same scored seconds. The alarm-budget experiment changes the validation operating point before evaluating event recall on held-out patients.

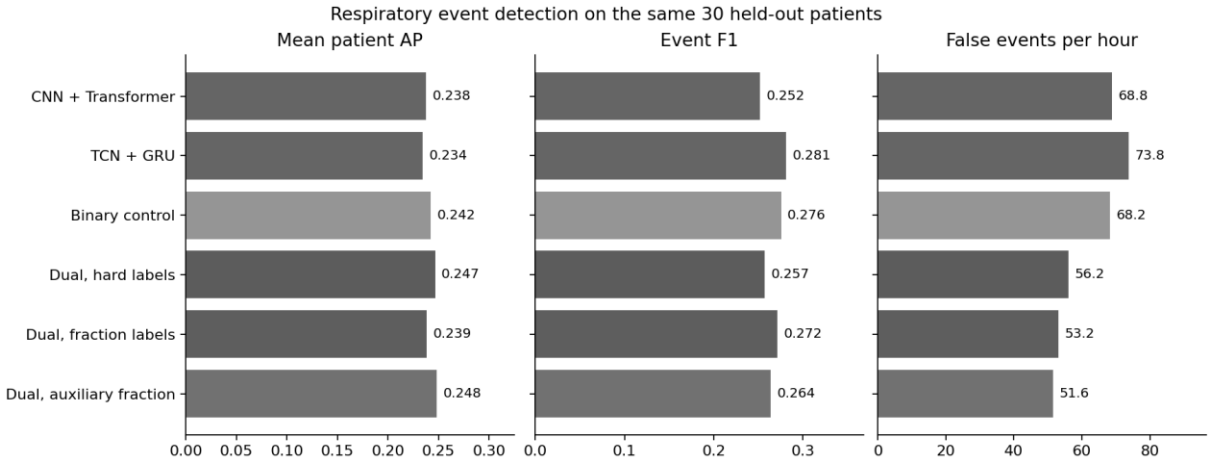


Figure A.27 Models

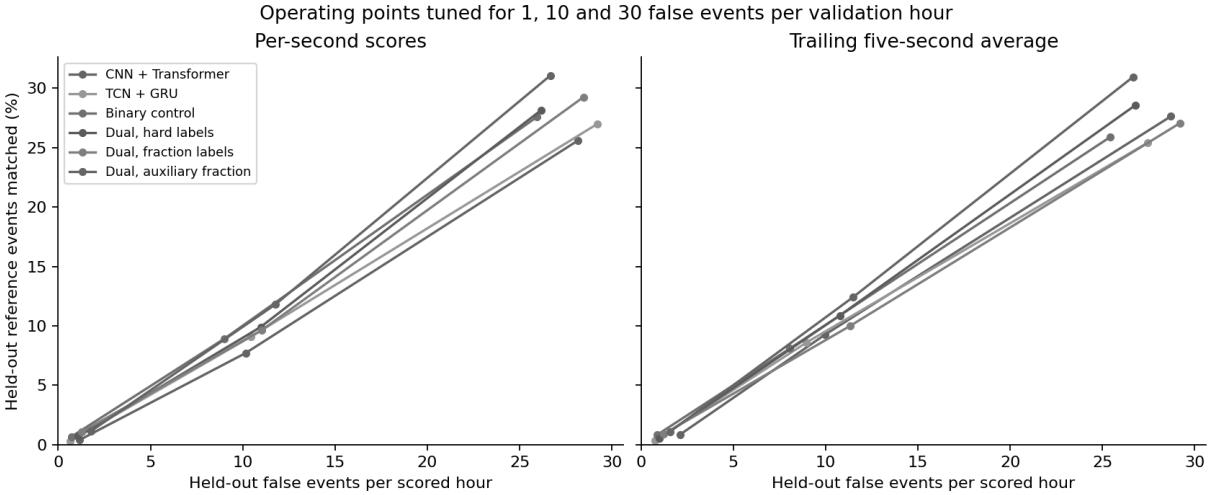


Figure A.28 Alert budgets

Source trials and gravity estimates

The sensor traces preserve acquisition order. Gravity-relative features summarize movement along and across estimated gravity even when the device axes rotate together.

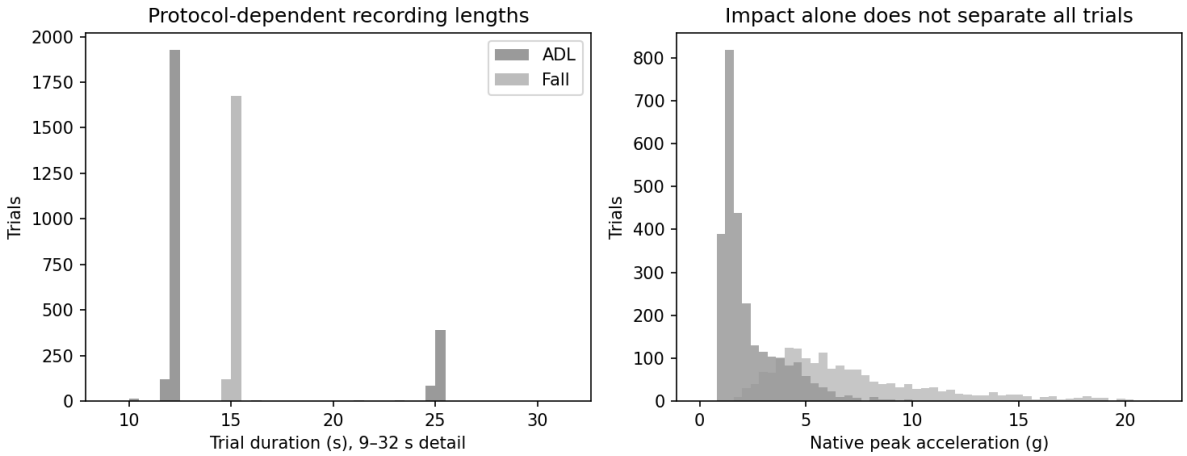


Figure A.29 Dataset profile

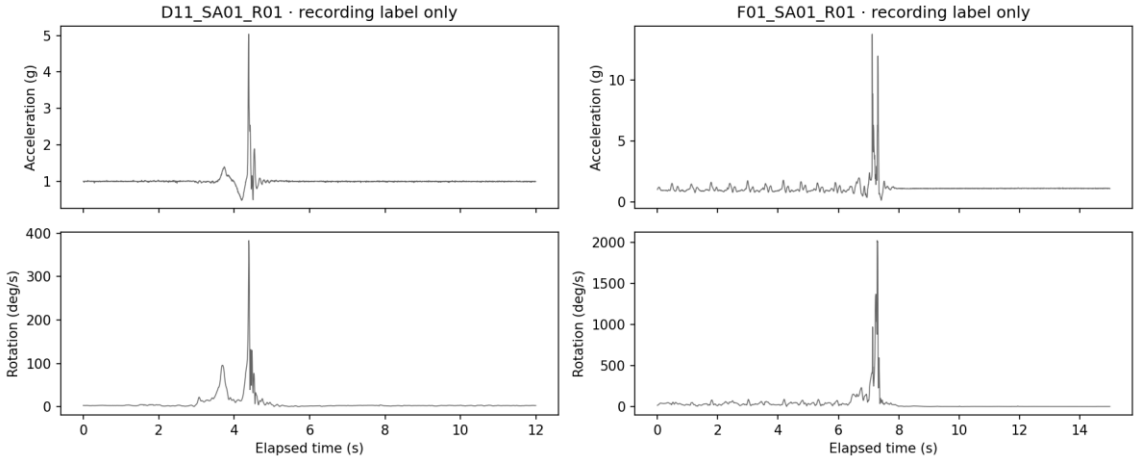


Figure A.30 Native signals

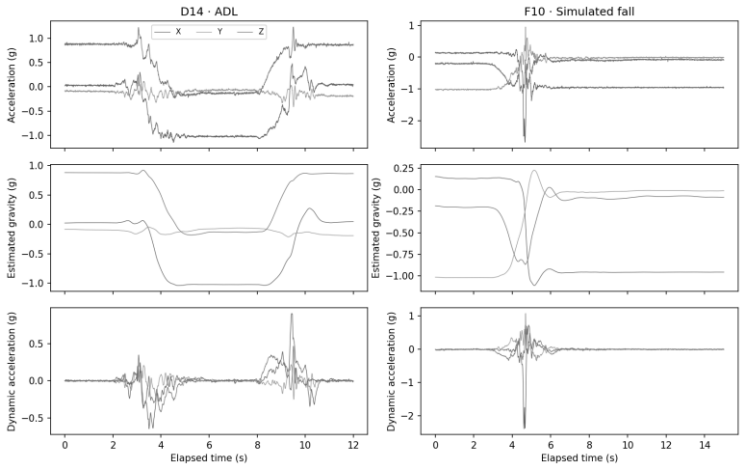


Figure A.31 Gravity components

Peak retention and feature ablations

The direct peak features use the native-rate samples. The filter audit separately measures how the smoothed path changes peak magnitude.

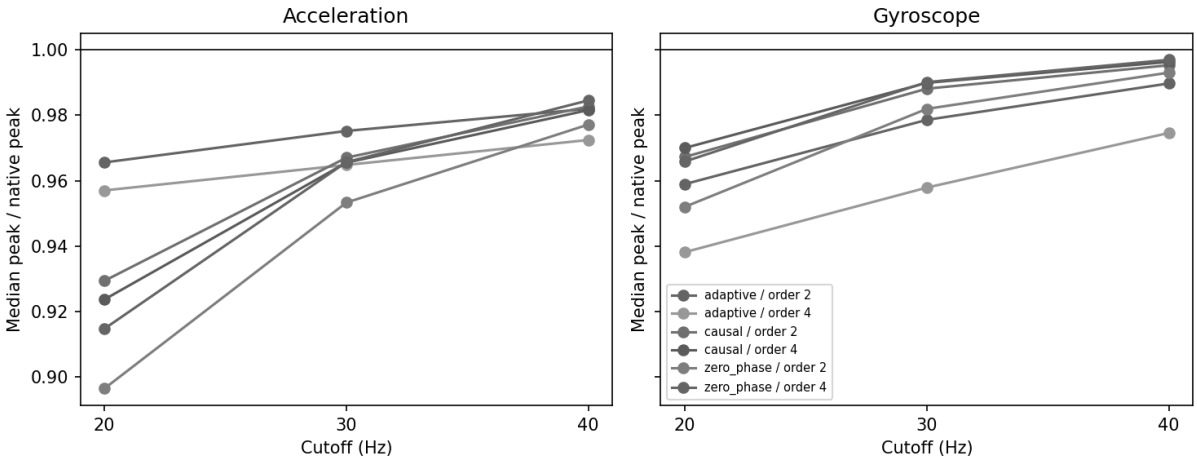


Figure A.32 Filter peak retention

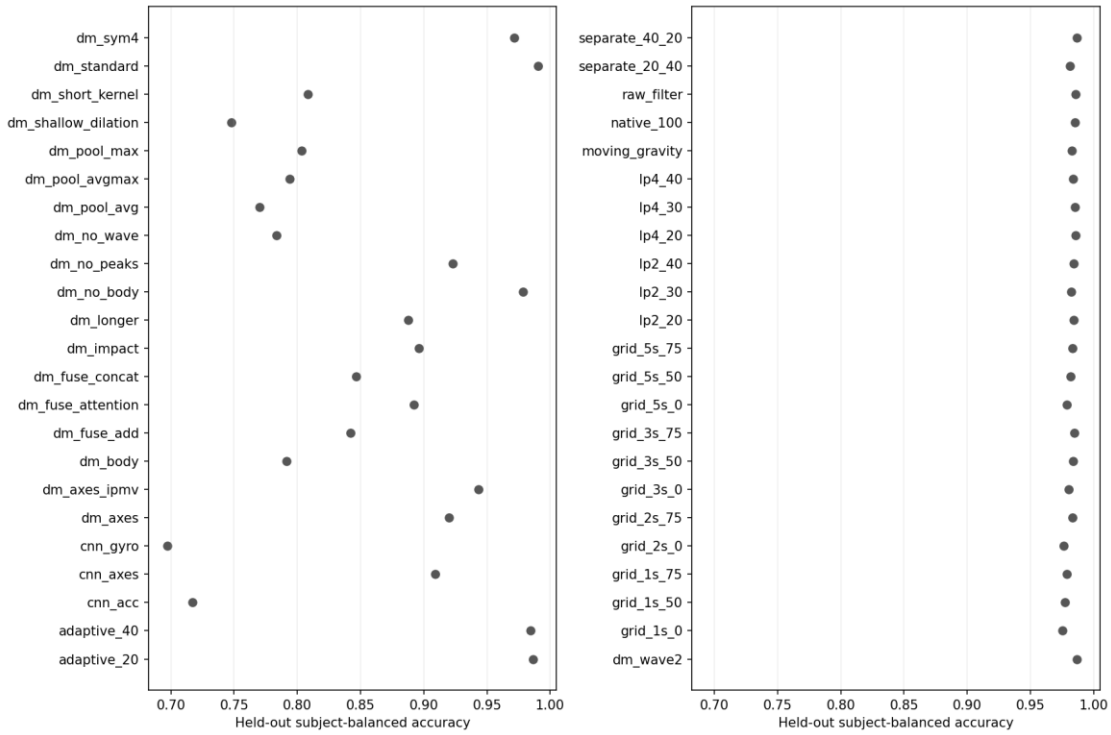


Figure A.33 Ablations

Architecture comparisons

The three-model view summarizes selected primary comparisons. The repeated-seed teacher-assisted experiment was conducted subsequently.

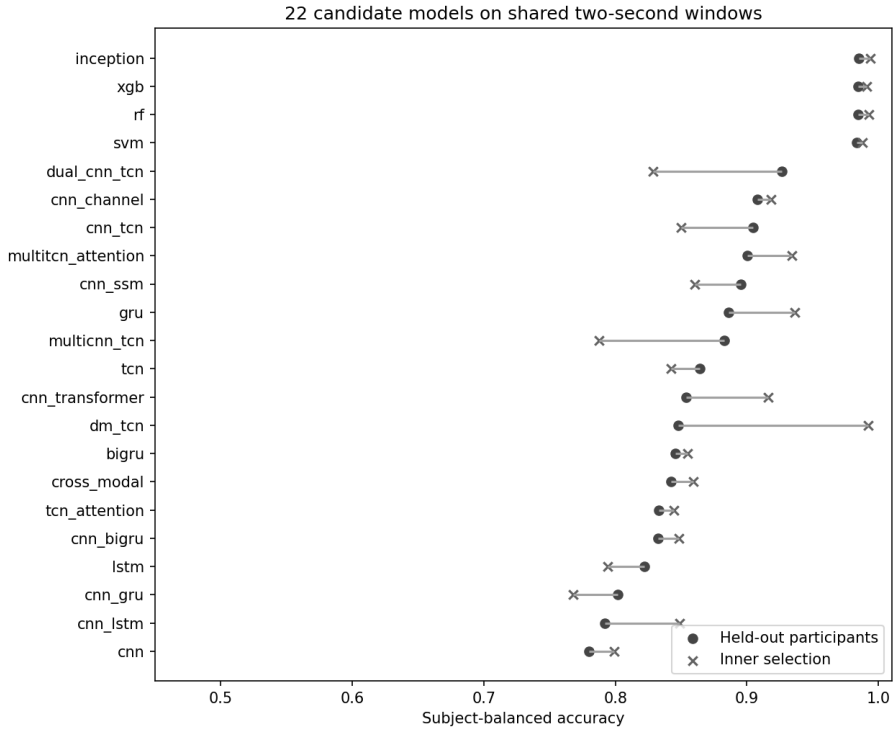


Figure A.34 Model comparison

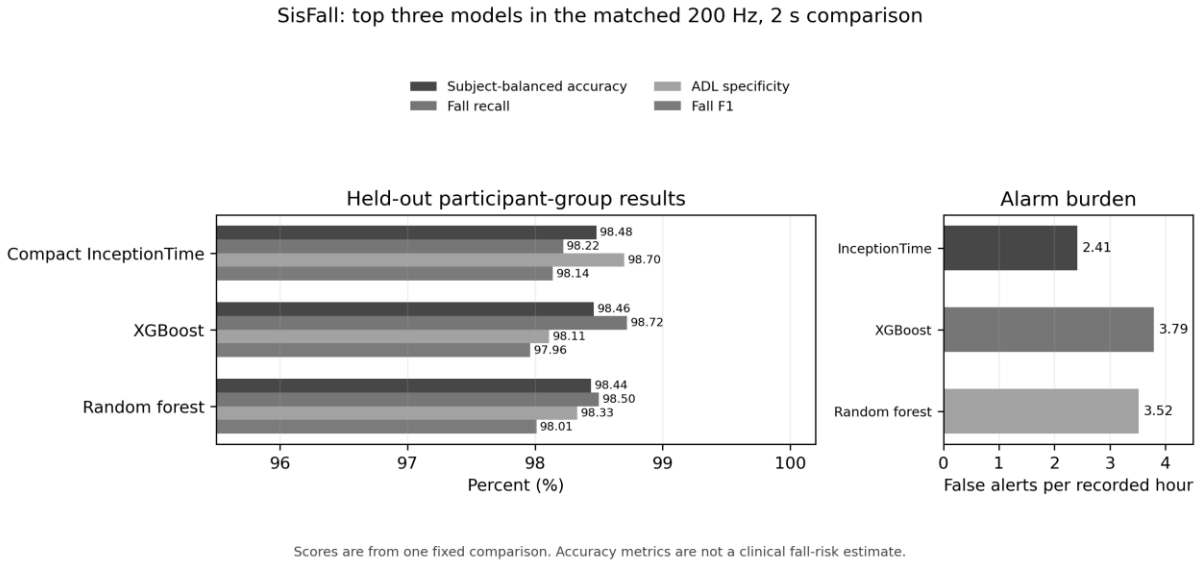


Figure A.35 Three primary model comparisons

Trial decisions and calibration

Whole-trial decisions use the maximum score among complete windows. These plots do not establish event-onset accuracy.

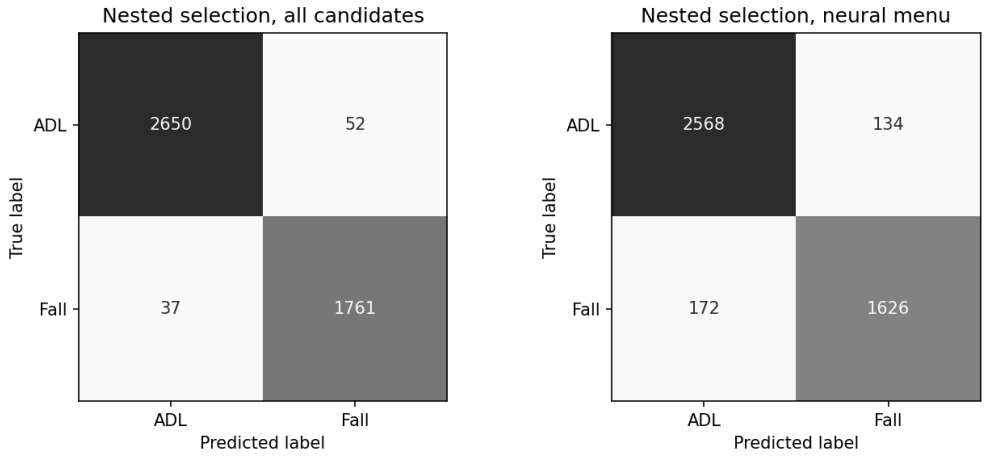


Figure A.36 Confusion

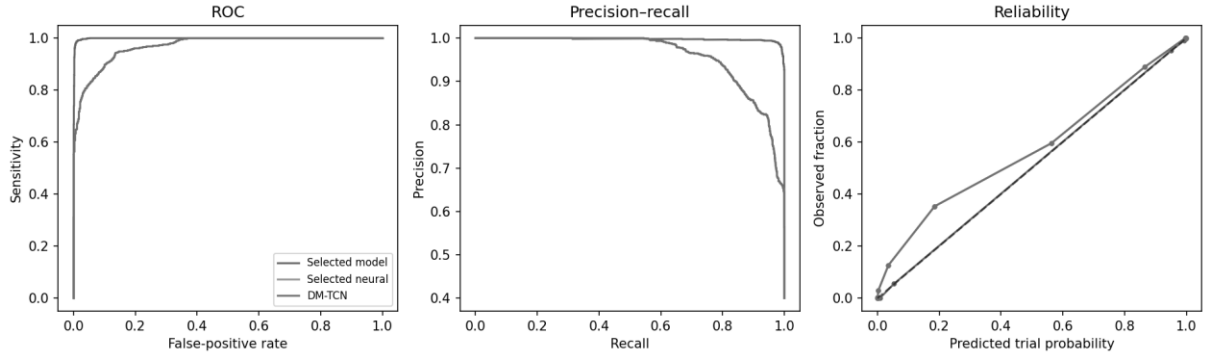


Figure A.37 Discrimination calibration

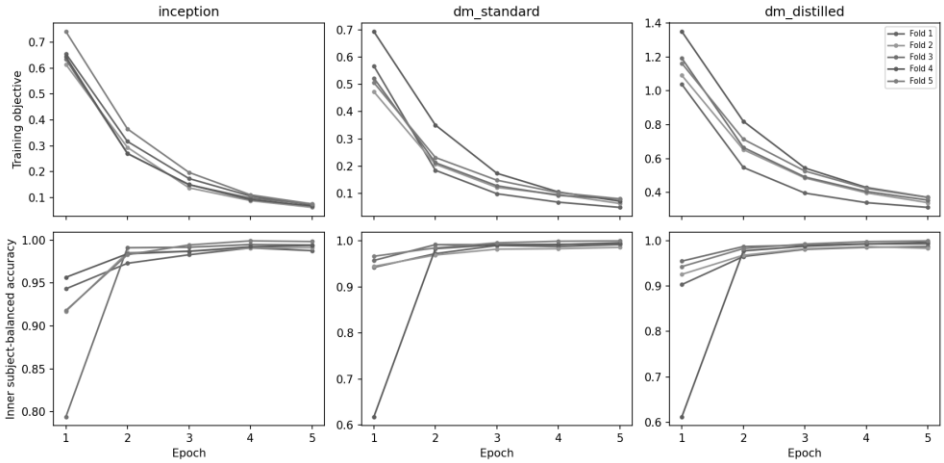


Figure A.38 Neural training

Initialization and transfer across sensor conditions

Repeated seeds reuse the same participants. Sensor perturbations are replay tests and provide no direct estimate of physical MPU-9250 accuracy.

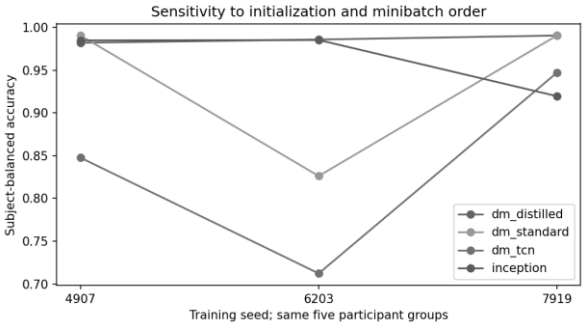


Figure A.39 Training-seed sensitivity

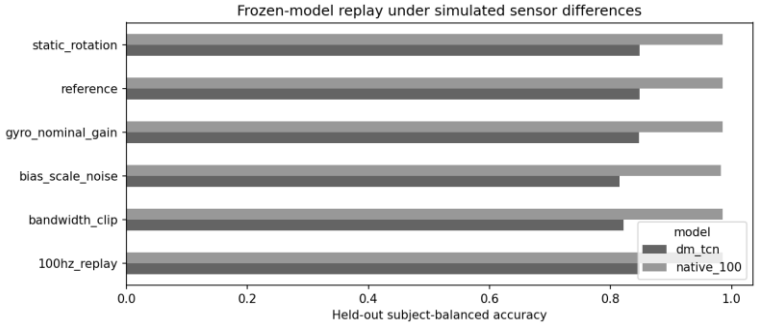


Figure A.40 Transfer

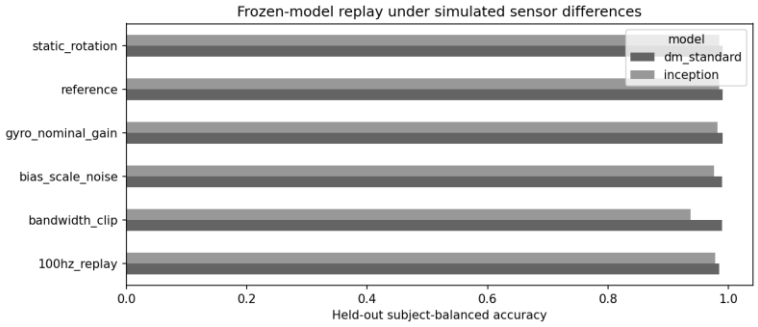


Figure A.41 Secondary transfer controls

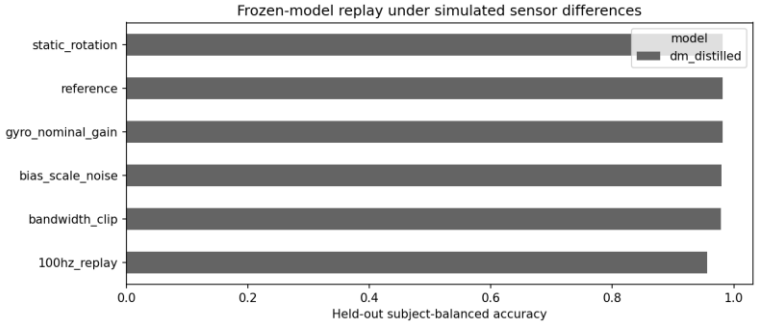


Figure A.42 Transfer of distilled DM-TCN

Cohort and signal decomposition

These panels compare observed conductance with algorithmic decompositions. WESAD supplies reference task labels but no independently measured tonic or phasic ground truth.

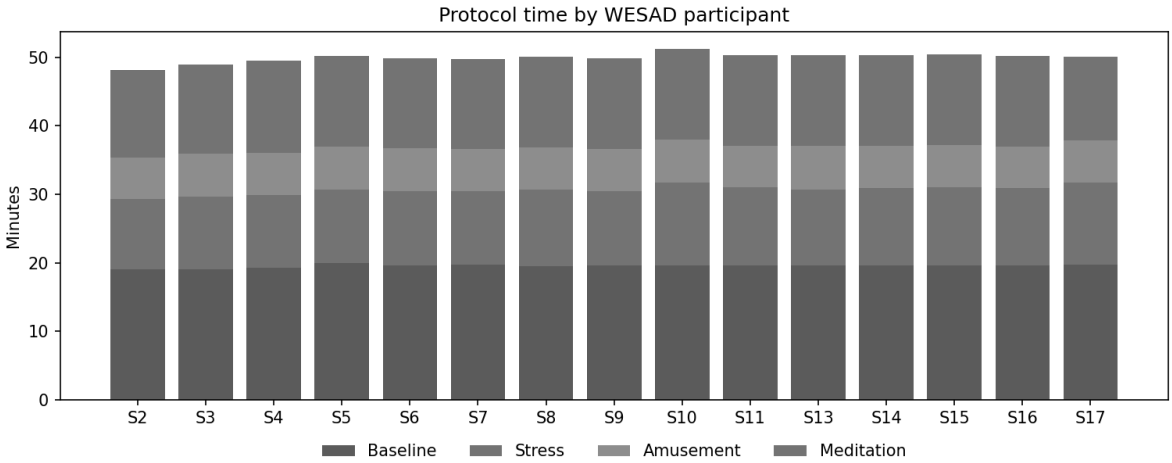


Figure A.43 Cohort conditions

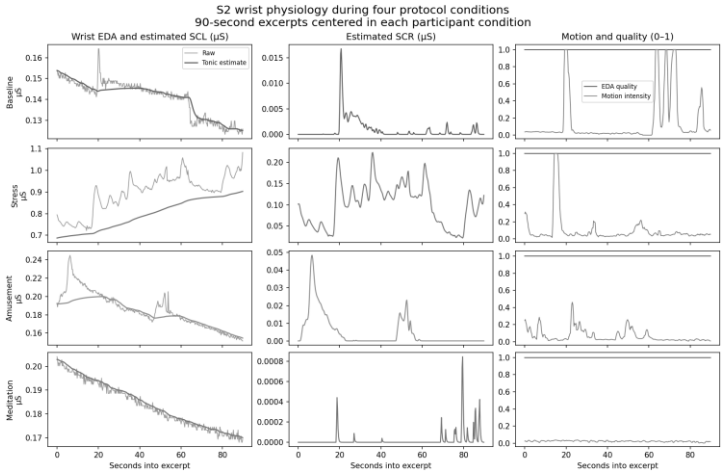


Figure A.44 Protocol examples

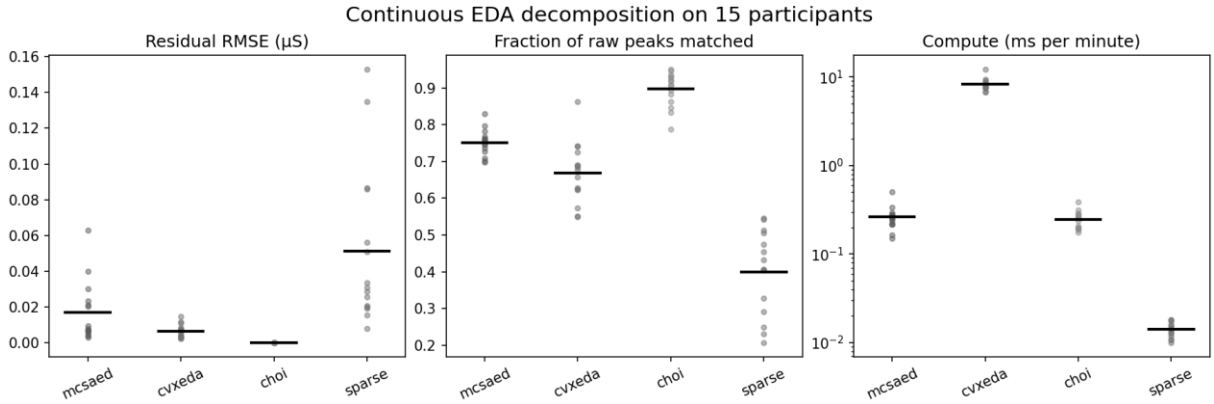


Figure A.45 Decomposition comparison

Effect of sensor availability

The wrist-only shortlist excludes accelerometry, temperature and pulse inputs. The broader architecture comparison is included to document how the original study's sensor assumptions change the ranking.

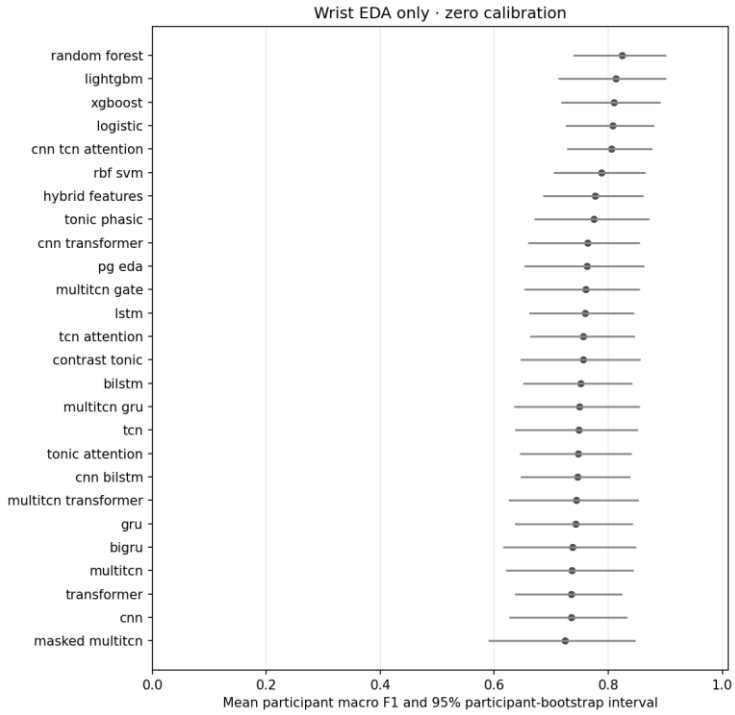


Figure A.46 Eda only architectures

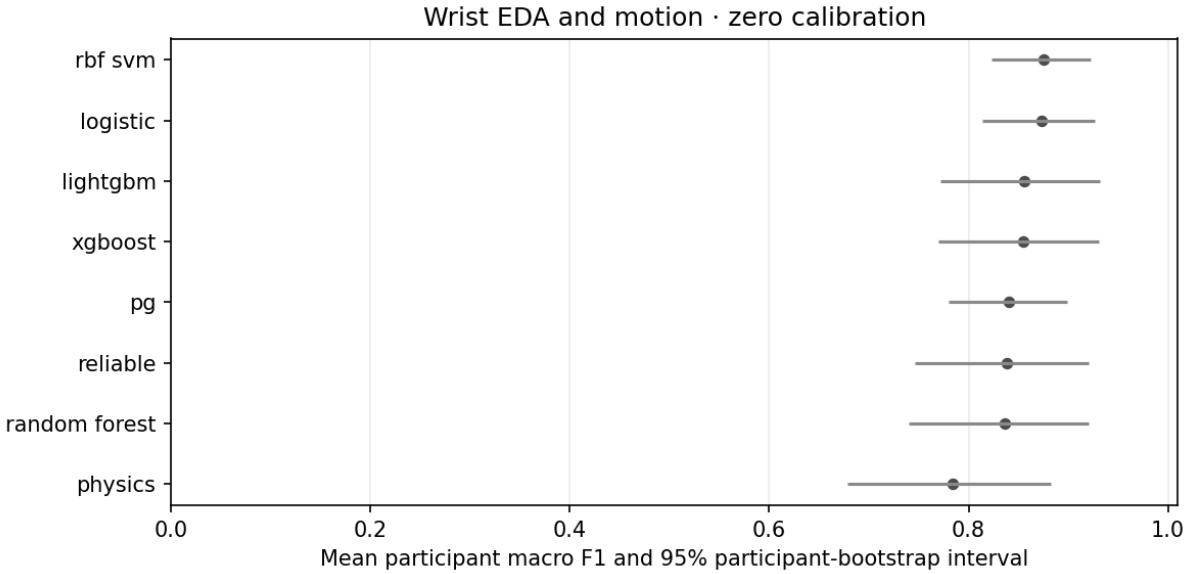


Figure A.47 Architectures

Ablations and protocol challenges

Matched ablations examine individual design choices. Protocol challenge results test how often laboratory conditions outside the binary training task receive a stress label.

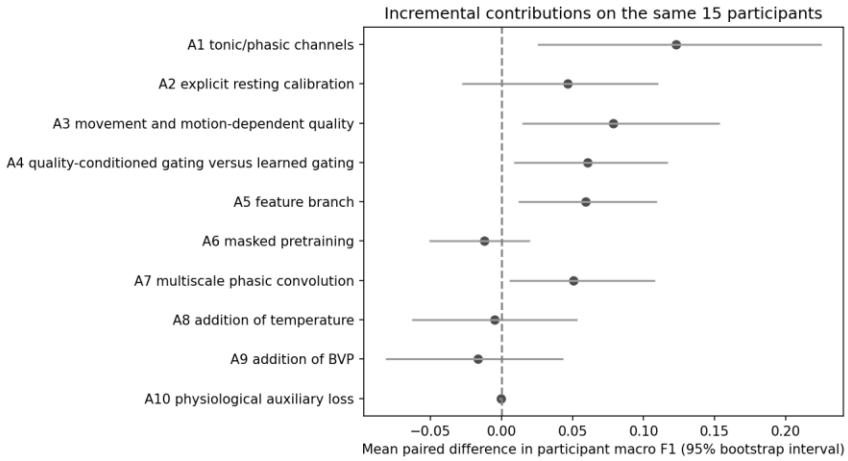


Figure A.48 Component ablations

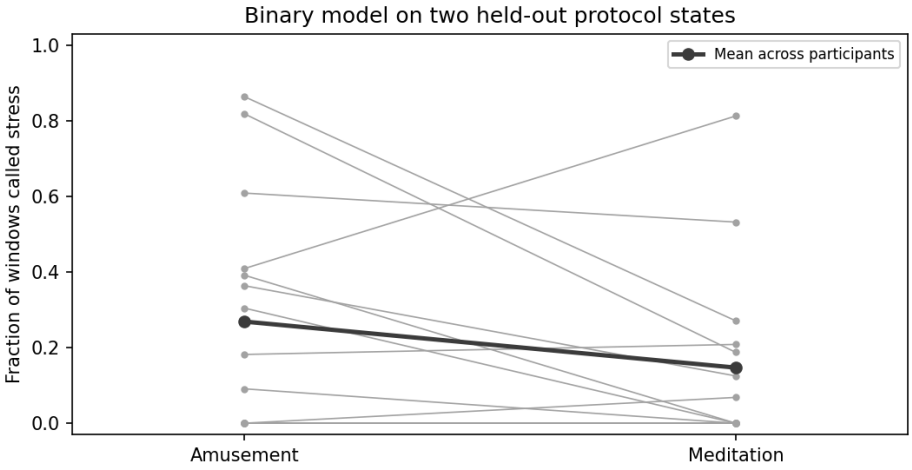


Figure A.49 Protocol challenge

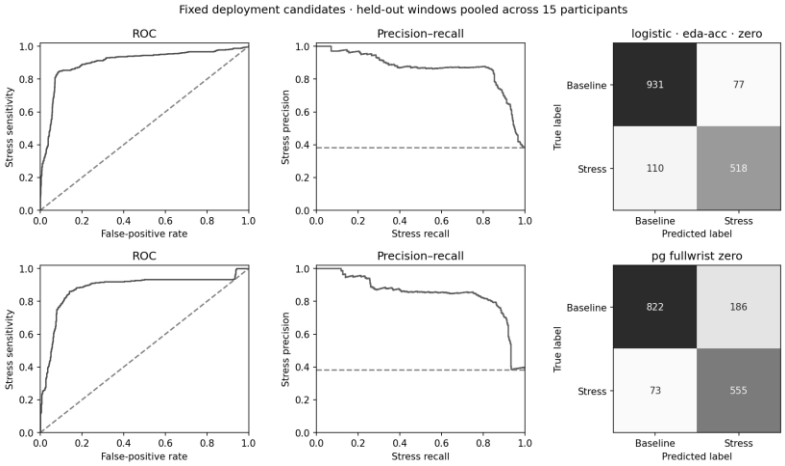


Figure A.50 Export candidate curves

Participant decisions with and without resting calibration

The zero-calibration protocol has no personal resting measurements. Subject-level plots expose poor individual outcomes that a pooled confusion matrix can obscure.

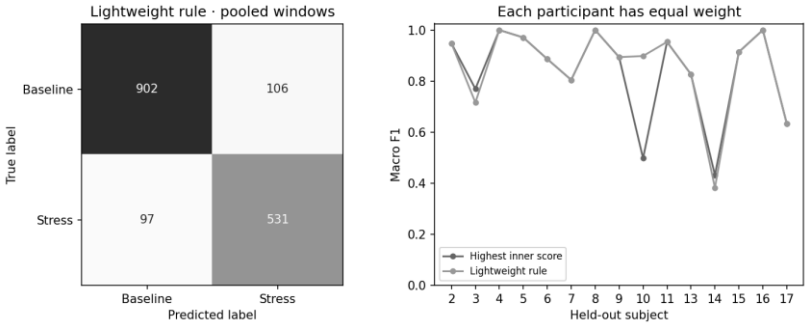


Figure A.51 Zero-calibration participant decisions

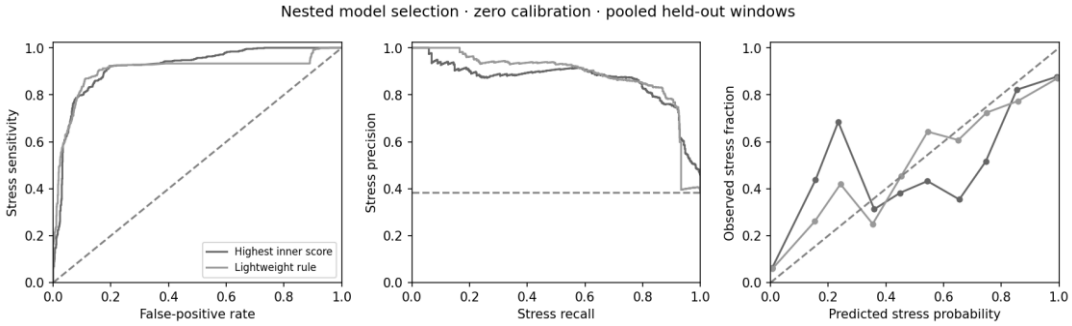


Figure A.52 Zero-calibration discrimination curves

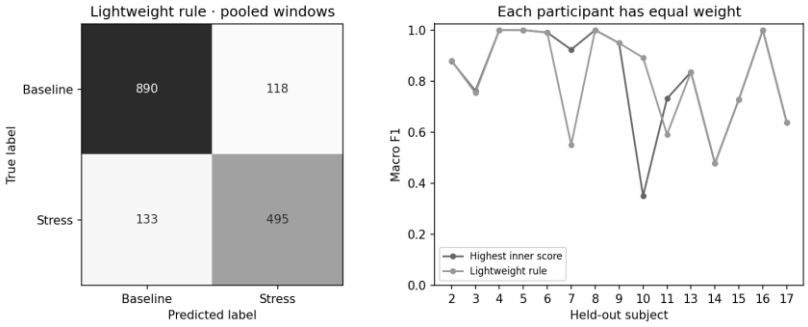


Figure A.53 Rest-calibrated participant decisions

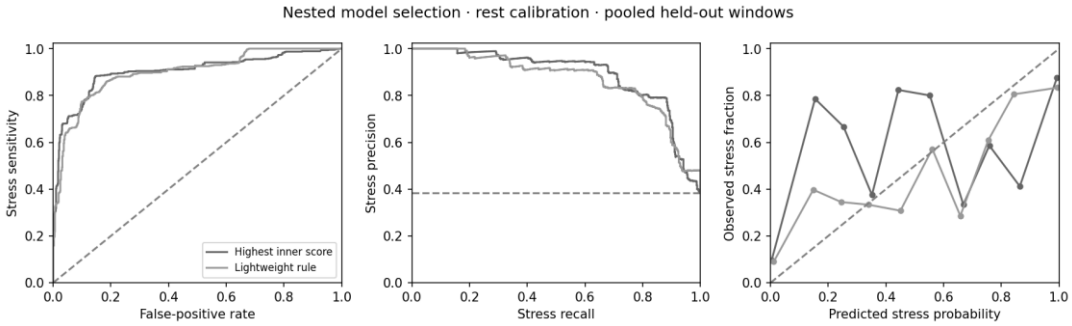


Figure A.54 Rest-calibrated discrimination curves

Neural verification

The audit covers the neural experiments, including experimental component-learning variants. The selected EDA-only CNN uses the causal MC-SAED components described in Chapter 3.

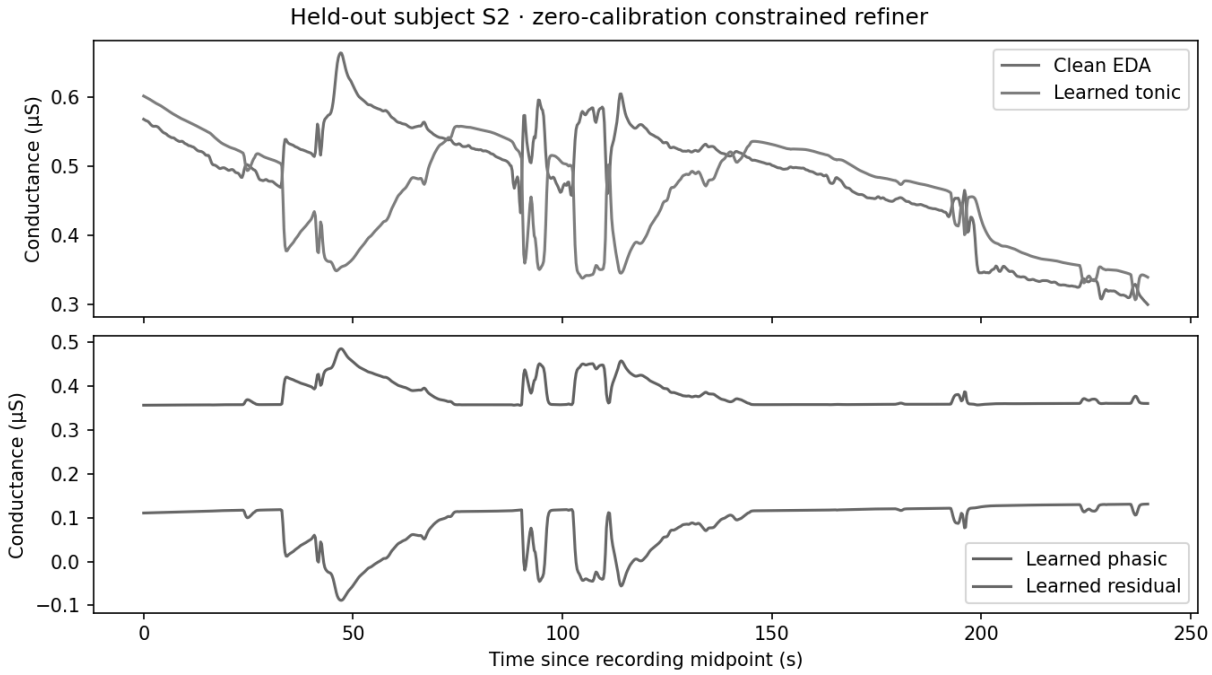


Figure A.55 Learned decomposition

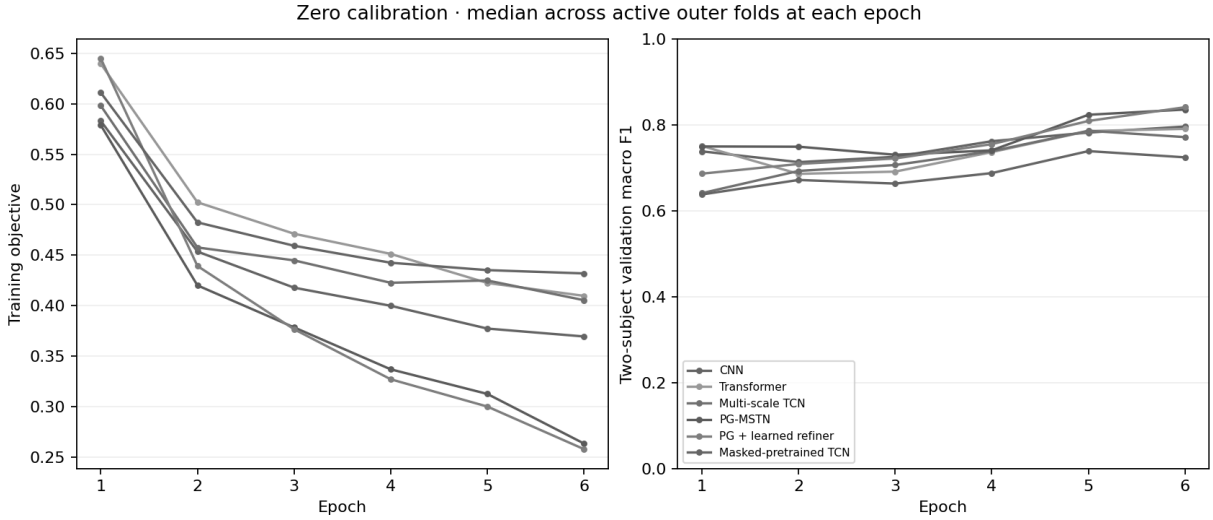


Figure A.56 Learning curves

Appendix B

Reproducibility and validation record

Experiment records pair each trained model with its participant partitions, normalization statistics and evaluation unit. The visual archive in Appendix A includes all 56 distinct saved result figures; duplicate image files and matched PNG/PDF versions are represented once.

Table B.1 Evidence needed to verify the selected pipelines

Module	**Verification record**	**Independent unit**
ECG	lvh_distill_v1 / final.csv	Patient and ECG
PPG	osasud_v1 / comparison.csv	Patient and event
IMU	sisfall_v1 / replicate_seeds.csv	Subject and trial
EDA	wesad_loso_v1 / all_models.csv	Participant

For ECG, filter the final table to prep_minimal and the 90% target-specificity operating point. Use wearable_window_metrics.json for the single-record confusion counts. For PPG, compute matched intervals divided by reference intervals, and unmatched predictions divided by scored hours.

For IMU, select dm_distilled and average the three subject-balanced accuracy entries. For EDA, select neural/cnn_rest and inspect participant-averaged macro F1. Those rows support Tables 4.1–4.4 alongside the original comparisons.

Physical validation should version the sensor settings, hardware calibration and signal-quality rules together with the model. Changes to channel position or input scaling can invalidate a previously measured operating threshold.